FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

AMCS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

ASSIGNMENT

SEMESTER 202601

Programme (Year & Group)	:	DCS1S3 G2
Tutor Name	:	MISS NG YI PEI
Tutorial Group	:	Group 2
Date Submitted	:	20/4/2026

No	Name	Student ID	Marks			Total Marks
			Section A	Section B	Section C	
1	TEO YAN QI	25WMD03207				
2	SIM YEN YUE	25WMD03006				
3	TAN YUN NI	25WMD09270				
4	KOH YAN CHENG	25WMD03680				

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AMCS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

Diploma in [Computer Science (DCS)]

Year 1 Semester 3

[Tutorial Group: Group 2]

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We declare that this assignment is free from all forms of plagiarism and for all intents and purposes is my own properly derived work.

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3	TAN YUN NI	25WMD09270		19/4/2026
4	KOH YAN CHENG	25WMD03680		19/4/2026

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Year 1 Semester 3

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TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Group Contract Form

Faculty	:	Faculty of Computing and Information Technology
Programme	:	Diploma in Computer Science
Year of Study	:	Year 1
Semester	:	Semester 3
Course Title	:	Object-Oriented Programming Techniques
Course Code	:	AMCS2204
Group Assignment / Project Title	:	Car Rental Management System
Name of Tutor	:	Miss Ng Yi Pei
Tutorial Group	:	Group 2

Student Name	Student ID	Role(s)/Task(s) Assigned	Date	Signature (DO NOT accept Digital Signature, please provide hand signature in image)
TEO YAN QI	25WMD03207	Return and Pickup Management	19/4/2026	
SIM YEN YUE	25WMD03006	Fleet and Vehicle Management	19/4/2026	
TAN YUN NI	25WMD09270	User and Profile Management	19/4/2026	
KOH YAN CHENG	25WMD03680	Booking and Reservation Management	19/4/2026	

Assignment Assessment Rubrics

No	Student Name	Student ID	Final Marks
1	TEO YAN QI	25WMD03207	
2	SIM YEN YUE	25WMD03006	
3	TAN YUN NI	25WMD09270	
4	KOH YAN CHENG	25WMD03680	

Programme of Study	D (CS1S3)
Tutorial Group	Group (2)

CLO 2	Demonstrate an object-oriented program using appropriate programming fundamentals with regards to arrays, methods and exception handling. (P4, PLO3)

Section A: Program Output & Correctness - Individual Marks (25 Marks)

Task	Weight	Max. Marks	Excellent	Good	Average	Poor	Very Poor	Scores (Point x Weight)	Remarks
			5	4	3	2	1		
Completeness of the program	**2.0**	**10**	Completed 100% of the functional requirements. All operations are implemented correctly.	Completed all of the functional requirements but implemented with minor errors.	Completed most of the functional requirements but implemented with errors.	Completed < 50% of the functional requirement and < 50% of the operations implemented correctly.	Did not complete any of the functional requirement at all	**No** / **Marks**: 1, 2, 3, 4	
Exception Handling	**1.0**	**5**	Output is correct with comprehensive	Output is correct with appropriate exception handling.	Output is correct with little exception handling.	Output is somewhat correct with little proper	Output is totally wrong with no proper exception handling.	**No** / **Marks**: 1	

			exception handling.			exception handling.		**2** **3** **4**	
Design of the Output	**1.0**	**5**	Excellent formatted output. Program displays more than expected	Good formatted output and program displays as expected.	Appropriate output design and display as expected.	Poorly designed output.	Minimum to no formatted output with messy display.	**No** / **Marks**: **1**, **2**, **3**, **4**	
Program Code Quality & Standard	**1.0**	**5**	All identifiers strictly conform to the standard Java naming convention and all are meaningful	Most of the identifiers conform to the standard Java naming convention and are meaningful	Some of the identifiers conform to the standard Java naming convention and some are meaningful	Limited numbers of the identifiers conform to the standard Java naming convention or are barely meaningful.	All of the identifiers conform to the standard Java naming convention or not meaningful	**No** / **Marks**: **1**, **2**, **3**, **4**	
						Total Marks (25 marks):		**1** **2** **3** **4**	

CLO 3	Analyse the concepts of encapsulation, inheritance and polymorphism based on programming problems. (C4, PLO2)								
Section B: Object-Oriented Concept - Group Marks (45 Marks)									
Task	**Weight**	**Max. Marks**	**Excellent**	**Good**	**Average**	**Poor**	**Very Poor**	**Scores (Point x Weight)**	**Remarks**
			5	**4**	**3**	**2**	**1**		
Class Implementation	**1.0**	**5m**	Identified the proper classes, methods and attributes to solve particular problems.	Identified appropriate classes, methods and attributes to solve particular problems with minor errors.	Able to identify only some classes, methods and attributes to solve particular problems.	Inappropriate classes, methods and attributes are used.	Unable to show understanding on the usage of classes, methods and attributes.		
Object Implementation	**1.0**	**5m**	Able to create a structure of objects collaborating among themselves to carry out tasks properly.	Able to create a structure of object collaborating among themselves to carried out task properly with little minor mistakes	Structure of collaborating object created with some mistakes	Structure of collaborating objects created is barely correct.	Wrong structure of object collaboration like one object doing everything itself.		
Abstraction	**1.0**	**5m**	Define the class at the proper level of abstraction (abstract classes and methods).	Define the class at the proper level of abstraction with minor mistake	Classes defined at the proper level of abstractions with some mistakes	< 50% of correct abstraction is used	Lack of abstraction (abstract classes and methods).		

Encapsulation	**2.0**	**10m**	Completely correct implementation of encapsulation (private modifier, setter and getter methods).	Correct implementation of encapsulation with only some minor mistakes	Acceptable amount of correct implementation of encapsulation is applied in program	< 50% of correct implementation for encapsulation applied in program	Completely incorrect implementation of encapsulation (private modifier, setter and getter methods)		
Inheritance	**2.0**	**10m**	Completely correct use of inheritance (correct use of extends keyword, super & sub class)	Correct implementation of inheritance with only some minor mistakes	Acceptable amount of correct implementation of inheritance is applied in program	< 50% of correct implementation for inheritance applied in program	Completely incorrect use and implementation of inheritance		
Polymorphism	**2.0**	**10m**	Completely correct use and implementation of polymorphism. Methods toString() & equals() correctly overridden.	Correct implementation of polymorphism with only some minor mistakes	Acceptable amount of correct implementation of polymorphism is applied in program	< 50% of correct implementation for polymorphism applied in program	Completely incorrect use and implementation of polymorphism.		
							Total Marks (45 marks):		

CLO 3	Analyse the concepts of encapsulation, inheritance and polymorphism based on programming problems. (C4, PLO2)								
Section C: Object-Oriented Concept - Group Marks (30 Marks)									
Task	**Weight**	**Max. Marks**	**Excellent**	**Good**	**Average**	**Poor**	**Very Poor**	**Scores (Point x Weight)**	**Remarks**
			5	**4**	**3**	**2**	**1**		
Cohesion	**1.0**	**5m**	The code demonstrates exceptional cohesion with all modules, classes, or functions having a clear and single responsibility.	The code demonstrates good cohesion overall where most modules, classes, or functions have a clear and single responsibility	Some modules, classes, or functions have a clear and single responsibility, but others may be slightly ambiguous	Few modules, classes, or functions have a clear and single responsibility.	Modules, classes represent more than one entity and lack of clear responsibility. Poor class cohesion.		
Coupling	**1.0**	**5m**	Minimal or no direct dependency between modules, classes, or functions. Coupling is effectively managed, resulting in highly maintainable and modular code	Most modules, classes, or functions have minimal direct dependencies.	Some modules, classes, or functions have moderate direct dependencies.	Many modules, classes, or functions have significant direct dependencies.	Modules, classes, or functions have excessive and tightly coupled dependencies which make module be harder to reuse or test.		
Association, Aggregation and Composition	**2.0**	**10m**	Associations between classes are accurately identified and	Associations between classes are generally identified and	Some associations between classes may be missing	Several associations between classes are missing or	Completely incorrect use and implementation of association,		

			properly established.	properly established.	or improperly established.	improperly established.	aggregation and composition relationship.		
UML Class Diagram	**2.0**	**10m**	The UML class diagram demonstrates excellent understanding and implementation. All classes, attributes, and methods are accurately represented with appropriate visibility, data types, and associations.	Most classes, attributes, and methods are accurately represented with appropriate visibility, data types, and associations.	Some classes, attributes, and methods may be missing or inaccurately represented, with visibility, data types, or associations not fully defined.	Many classes, attributes, and methods are missing or inaccurately represented, with visibility, data types, or associations not adequately defined.	Classes, attributes, and methods are missing or inaccurately represented, with visibility, data types, or associations lacking clarity or definition.		
							Total Marks (30 marks):		

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1.0 Introduction

System Idea（Car Rental Management System）

Introduction

The **Car Rental Management System** is a software platform designed to automate and optimize the management of vehicle fleets, customer reservations, and rental return logistics. As the global transportation and travel industry becomes more complex, the requirement for precise coordination of mobile assets including Economy, Luxury, and RV vehicles is increasingly essential for business survival.

This system aims to improve the workflow of customer registration, vehicle availability checking, booking, and return processes. It ensures that vehicles such as Economy, Luxury, and RV cars can be managed properly without conflicts or double booking. By integrating all functions into one system using object-oriented programming concepts, the **Car Rental Management System** helps to reduce human errors, improve efficiency, and provide better service to customers. In conclusion, the **Car Rental Management System** offers a reliable and user-friendly solution for managing car rental operations.

1.1 Fleet and Vehicle Management

The Fleet and Vehicle Management module is responsible for managing all vehicles that the company has input in the system. It allows staffs to add new vehicles by entering details such as picture ,plate number, rental price, and vehicle type in a simple form.Moreover, the system provides a delete function to remove vehicles that are no longer in use. Category filters are also available to help staff organize vehicles based on types Economy, Luxury, and RV . This module ensures that all vehicle information is accurate and up-to-date.Staff can also check the vehicle condition, like if it needs to go for maintenance. Staff can choose between repair need , service require,and routine maintenance to describe and submit a report to the HQ for approval for action.

1.2 User and Profile Management

The User and Profile Management module ensures the security and personal data associated with the system's participants. Following a structured inheritance model, the system manages two primary roles: Customers and Staff. For customers, the module facilitates a Registration process where driving license numbers and contact details are securely stored to verify rental eligibility. For staff, the module provides specialized access levels to perform administrative tasks. By separating these roles through encapsulation, the system ensures that sensitive data is protected and that users only interact with the functions relevant to their specific permissions.

1.3 Booking and Reservation Operations

The Booking and Reservation module allows customers to rent vehicles easily. Users can search for available cars by selecting the start and end dates of the rental period. The system will then perform a real-time availability check to ensure that the selected vehicle is not already booked. Once a vehicle is selected, the system will generate a booking ID and calculate the estimated rental cost based on the duration and vehicle rate. Customers can also modify or cancel their bookings if needed. This module helps to organize the rental process and provides a smooth booking experience for users.

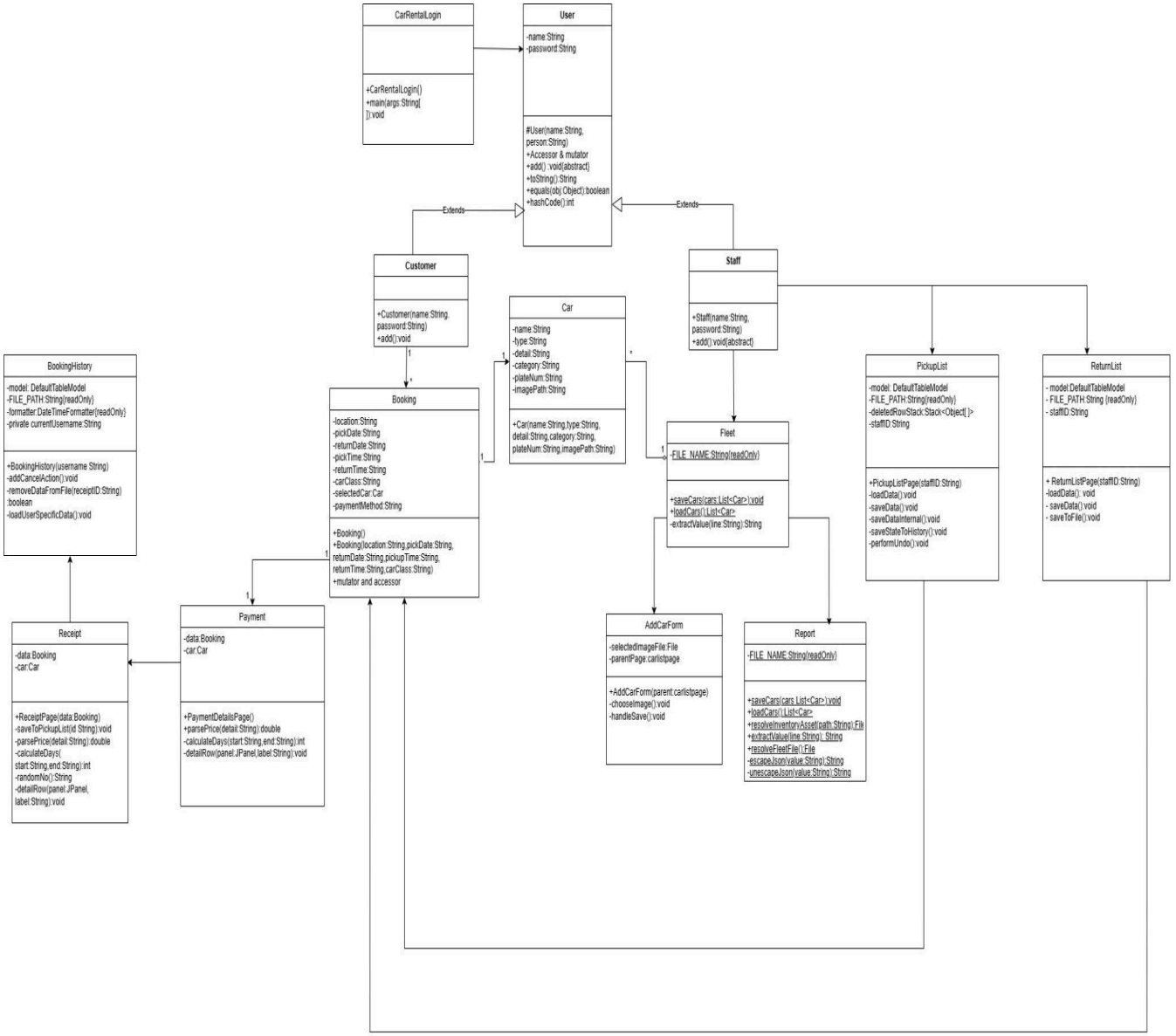
1.4 Return and Pickup Management

The Return Management module handles the process when a vehicle is returned. Staff will check the vehicle condition, such as whether it needs maintenance. Staff can choose between repair needs , service requirements, or routine maintenance to describe and submit a report to the HQ for approval for action. Moreover , this module allows staff to manage the status of each car when returning, like “Return” or “Late Return”. Furthermore, the Pickup Management Module handles the process when a vehicle is already picked up by the customer. All the staff need to do is check the customer's receipt code and help the customer to pick the car then the staff will enter the selected car plate to make it in "rented” condition.

2.0 Task Allocation

Student Name	Modules Handled	Description
TEO YAN QI	Return and Pickup Management	Pickup List, Pickup List ,Return List, Staff Mainpage,Car Rental Login (portal), Photo
SIM YEN YUE	Fleet and Vehicle Management	Car ,Fleet , AddCarForm ,Report ,Report Page,maintenance_report,fleetData ,fleet_inventory ,Photo
KOH YAN CHENG	Booking and Reservation Management	Booking Page, Choose Car Page, Payment Details Page, Choose Payment Page, Receipt Page, Booking History
TAN YUN NI	User and Profile Management	Cust, Customer ,Cust Login, Signup Page , Cust Mainpage,Staff, Staff, Staff Login ,User

3.0 UML Diagram

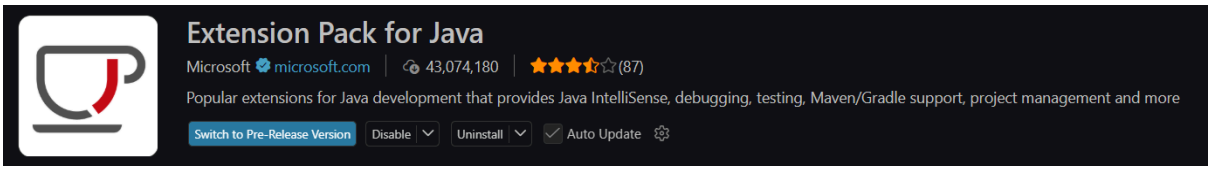


https://app.diagrams.net/?state=%7B%22ids%22:%5B%2214zCDRFWlsvRl1UySfTMI0g9Lfw-omCI4%22%5D,%22action%22:%22open%22,%22userId%22:%221033281159817726902 76%22,%22resourceKeys%22:%7B%7D%7D

4.0 System Workflows

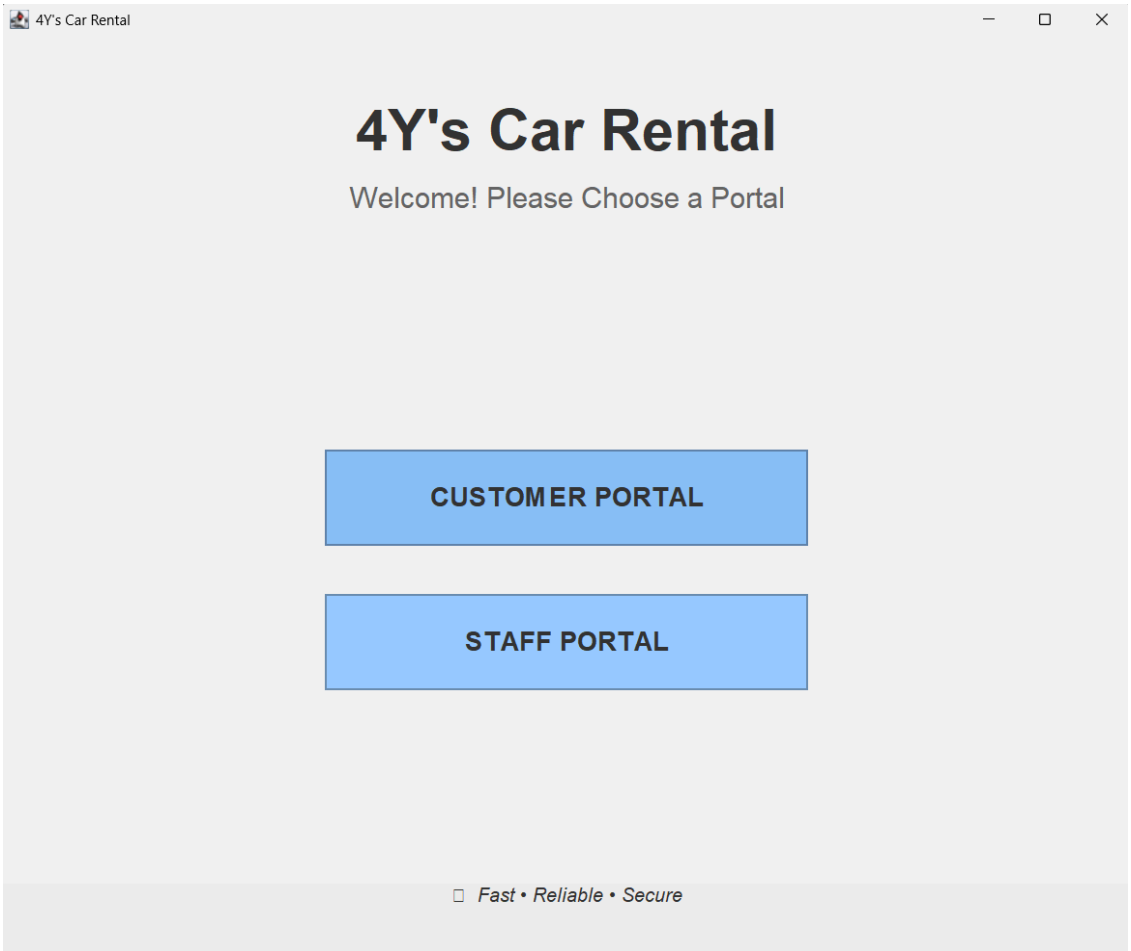
4.1 Main Program

Note : use **Visual Studio Code (terminal)** to run our system for the best result. Thank you.



//Debugger for Java, Extension Pack For Java,Gradle for Java,Maven For Java,Project Manager for Java,Test Runner For Java, Language support For Java

4.1.1 Main Page



The menu option include:

[1] Customer Portal

- Allow customers to log in with their name and password or sign up into the system.

[2] Staff Portal

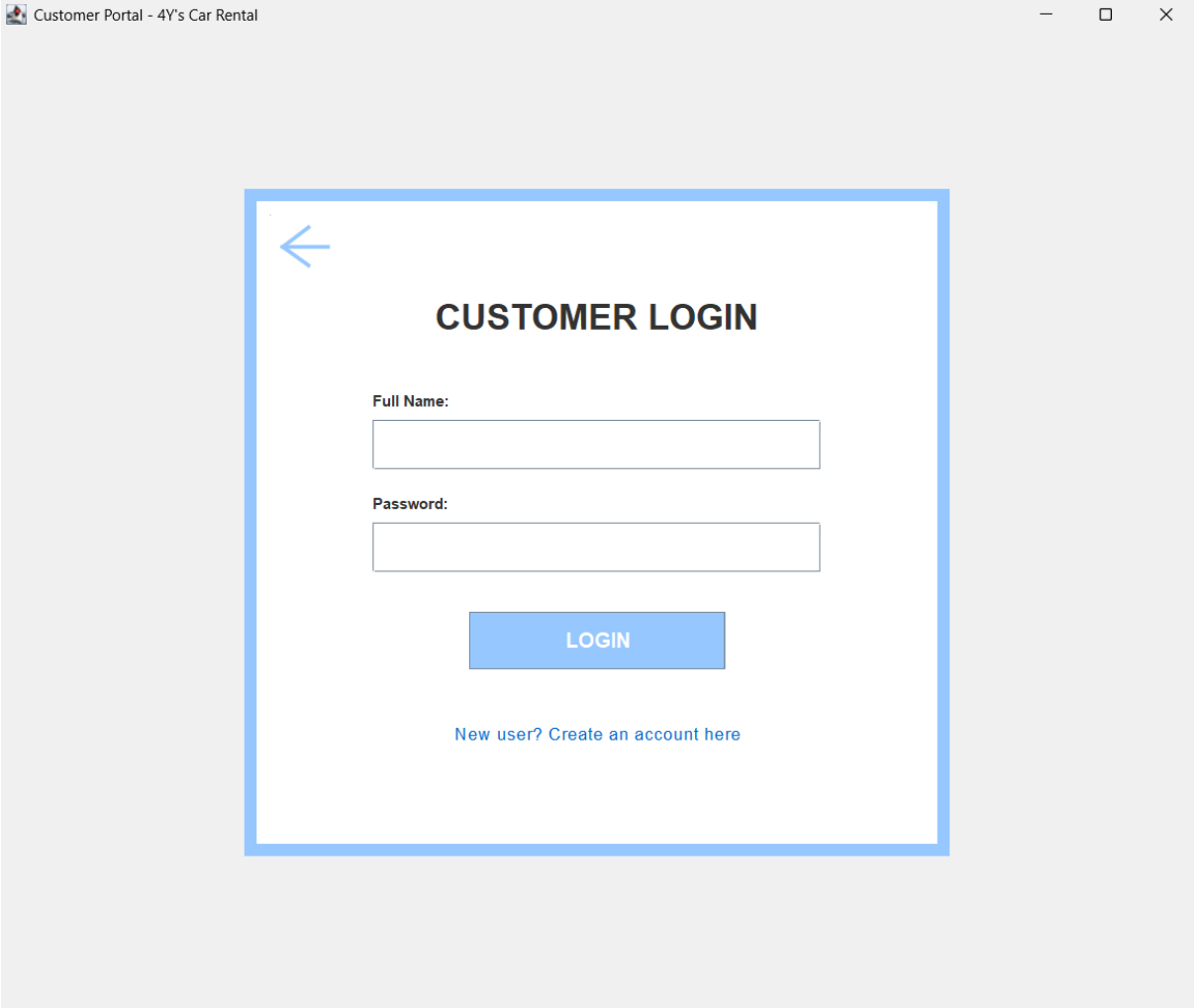
- Allow staff to login into the system with their staffID and the correct password.

Steps:

1. The system displays main menu options.
2. User selects role (Customer or Staff).
3. The system directs the user to the selected module.

4.1.2 Customer Sign Up & Login

4.1.2.1 Customer Login Page



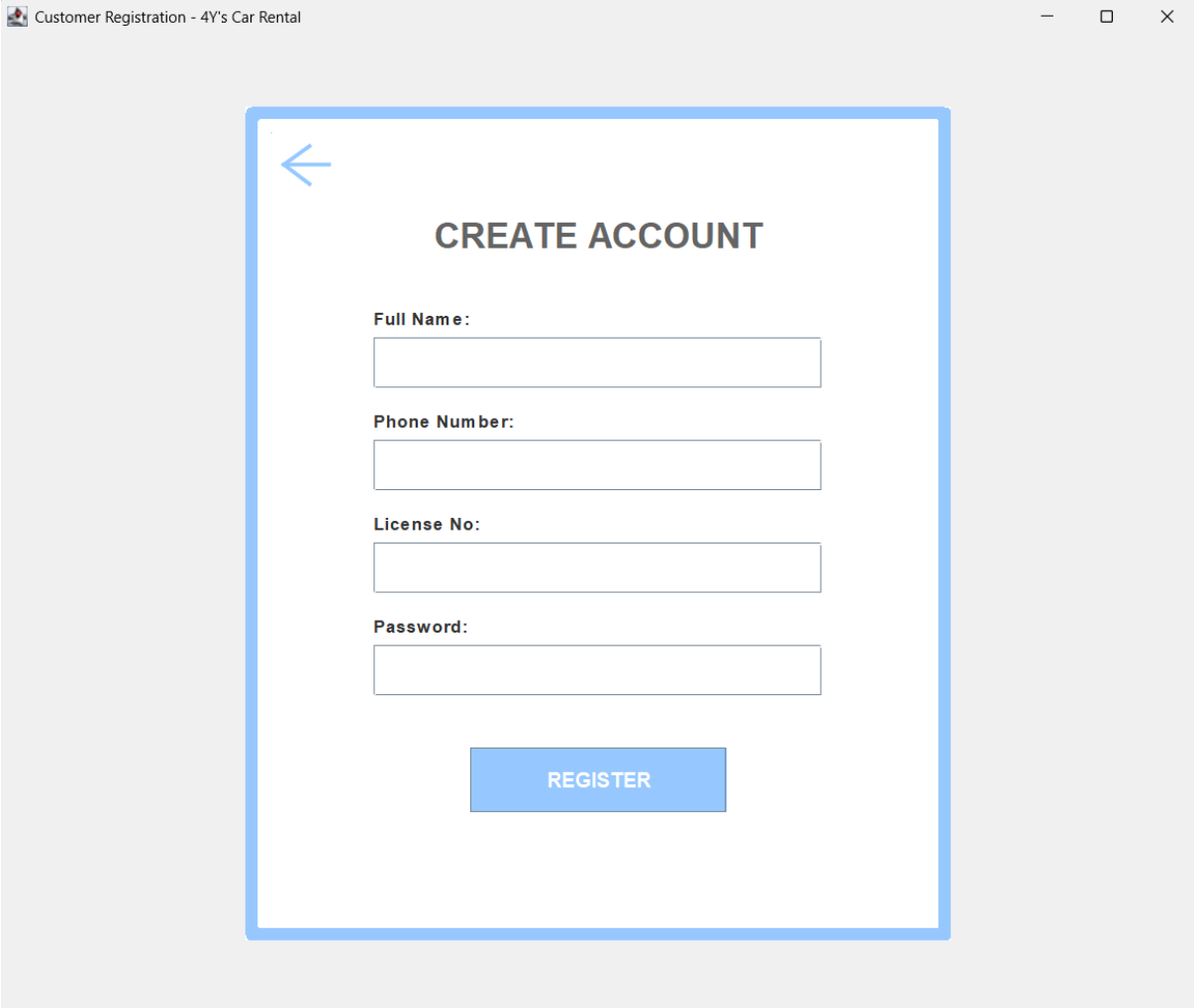
In the Customer Login Page, customers can enter their full name and password to login their account. If the new user does not have an account, they can click the blue sentences under the **Login** button to jump to the Sign Up Page.

Steps:

1. Customer selects “Login”.
2. Customer enters name and password.
3. The system verifies the information.
4. Customer clicks on the login button.
5. If correct, the customer will enter the booking page.
6. If incorrect, the system displays “Wrong Name or Password”.

7. If it is empty, the system displays “Invalid Name or Password”.

4.1.2.2 Customer Sign Up Page



In the Customer Sign Up Page, customers can enter their full name, phone number, license no and create a safe password to register a new account. If the customer already has an account, they can click the **Back** button in the upper left corner to go back to the Login Page.

Steps:

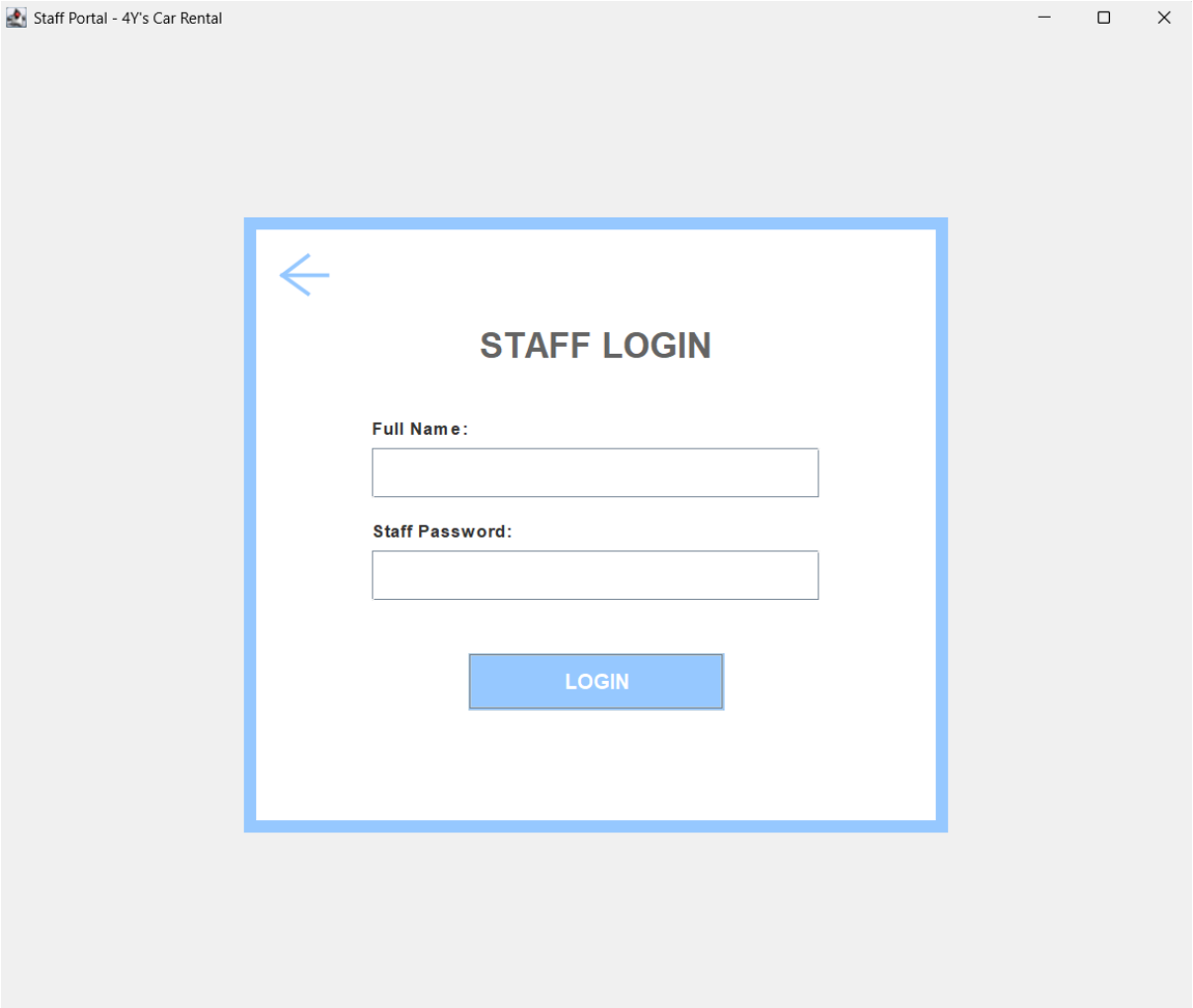
1. Customer selects " Sign up”.
2. The system prompts for name, IC number, phone number, driving license class, email and password.
3. Customer enters required details.
4. Customer clicks on the signup button.
5. The system saves the data.
6. “Registration Successful" message displayed then the customer enters the login page.

Notes:

- If it is empty, the system displays “Please complete all fields”.

- The phone number must be 10-11 digits only. If incorrect, the system will display an error message.
- The license number must be exactly 12 digits only. If incorrect, the system will display an error message.
- The password must be 6-11 characters long. If incorrect, the system will display an error message.

4.1.3 Staff Login

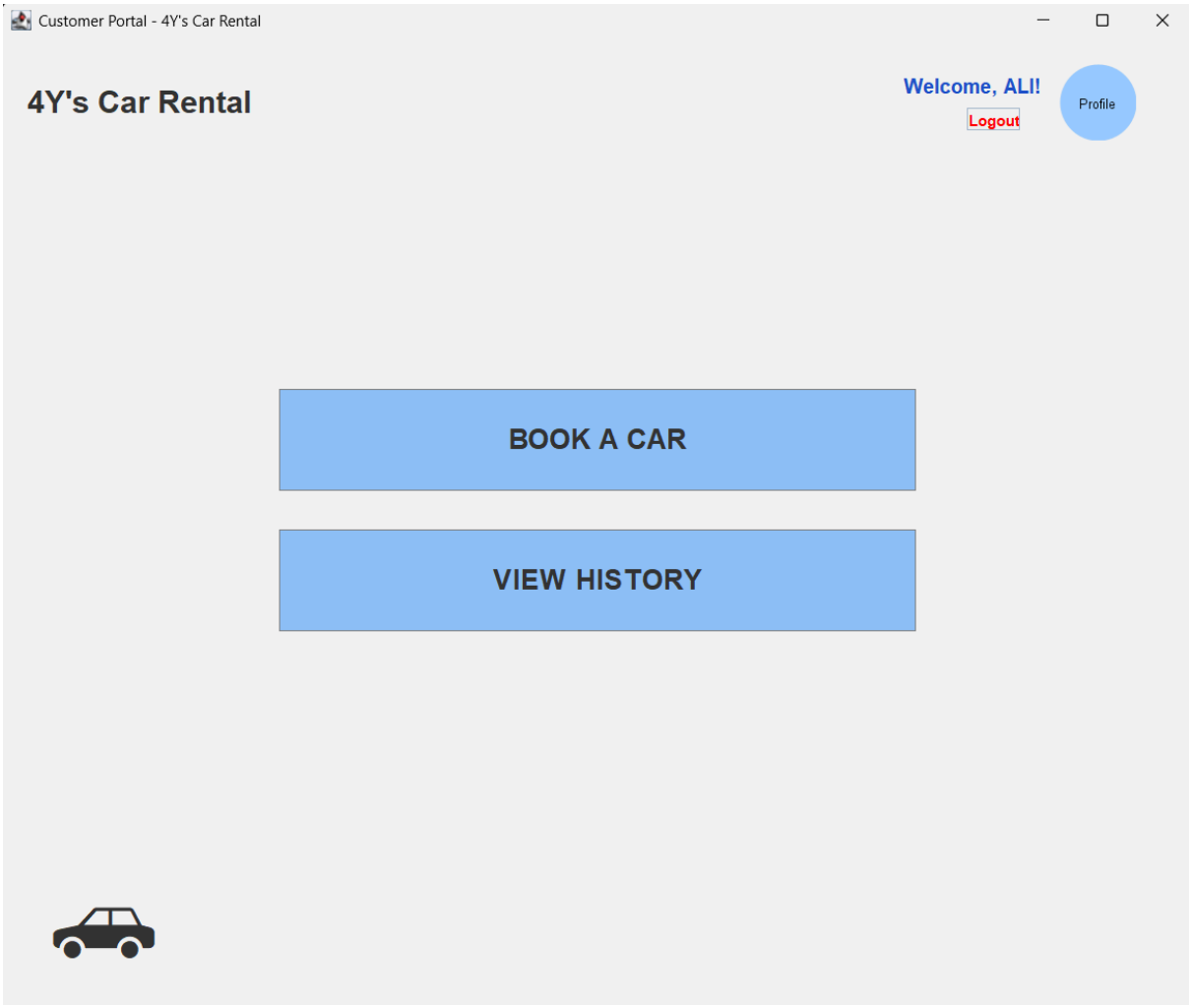


In the Staff Login Page, staff can enter their full name and password to login their account.

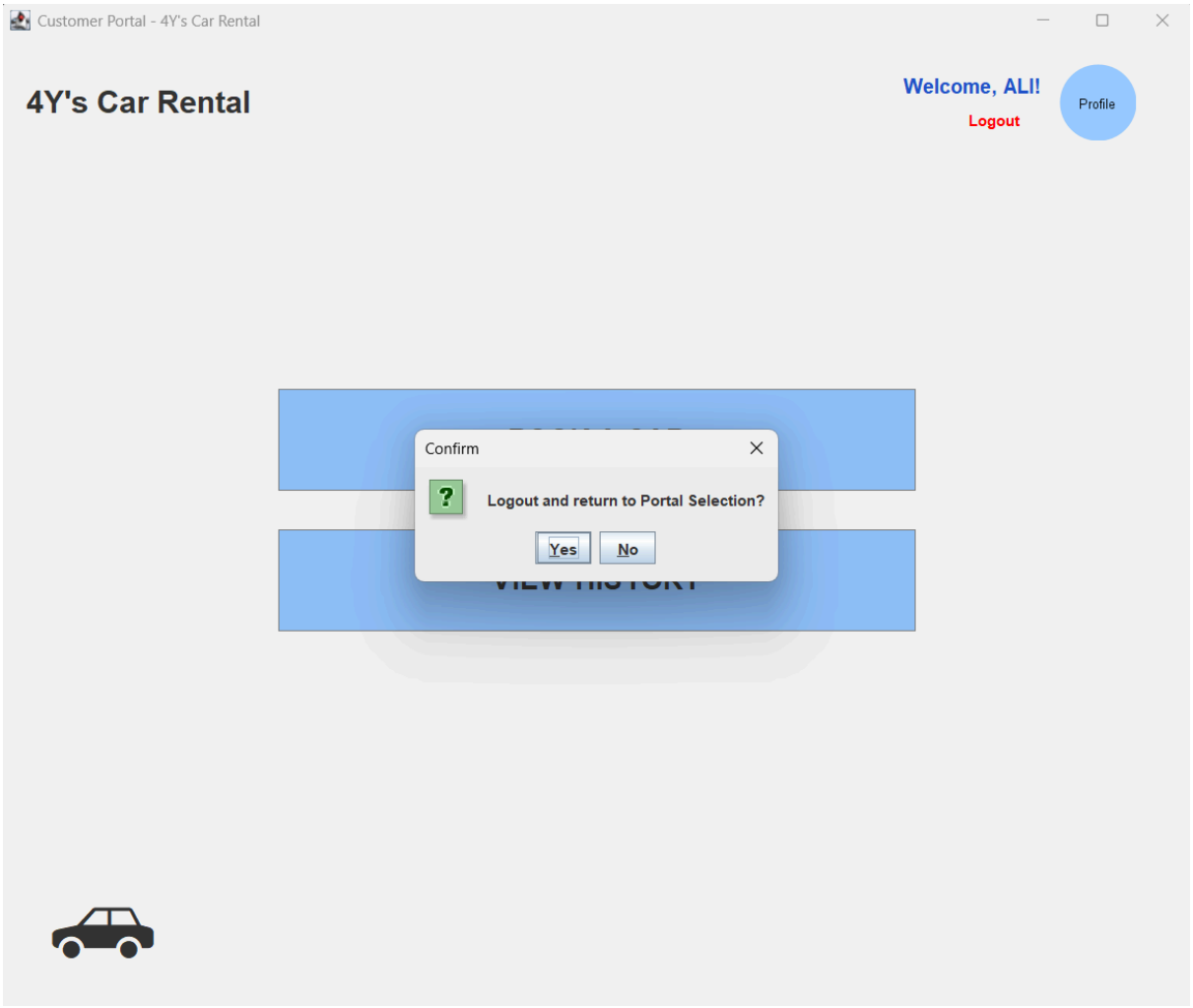
Steps:

1. Staff selects "Login".
2. Staff enters full name and password.
3. The system verifies the information.
4. If correct, staff enters the system.
5. If incorrect, the system displays an error message.
6. If it is empty, the system displays "Invalid Name or Password".

4.2 Customer Management



The Customer Main Page includes two options, which are **Book a Car** and **View History**. The upper right corner will display the "Welcome, (username)!" and the **logout** allows customer to logout their account. When the customer clicks the sentence, the system will display the confirmation message to confirm their action.



// confirmation message to logout account

The menu option include:

[1] Book a Car

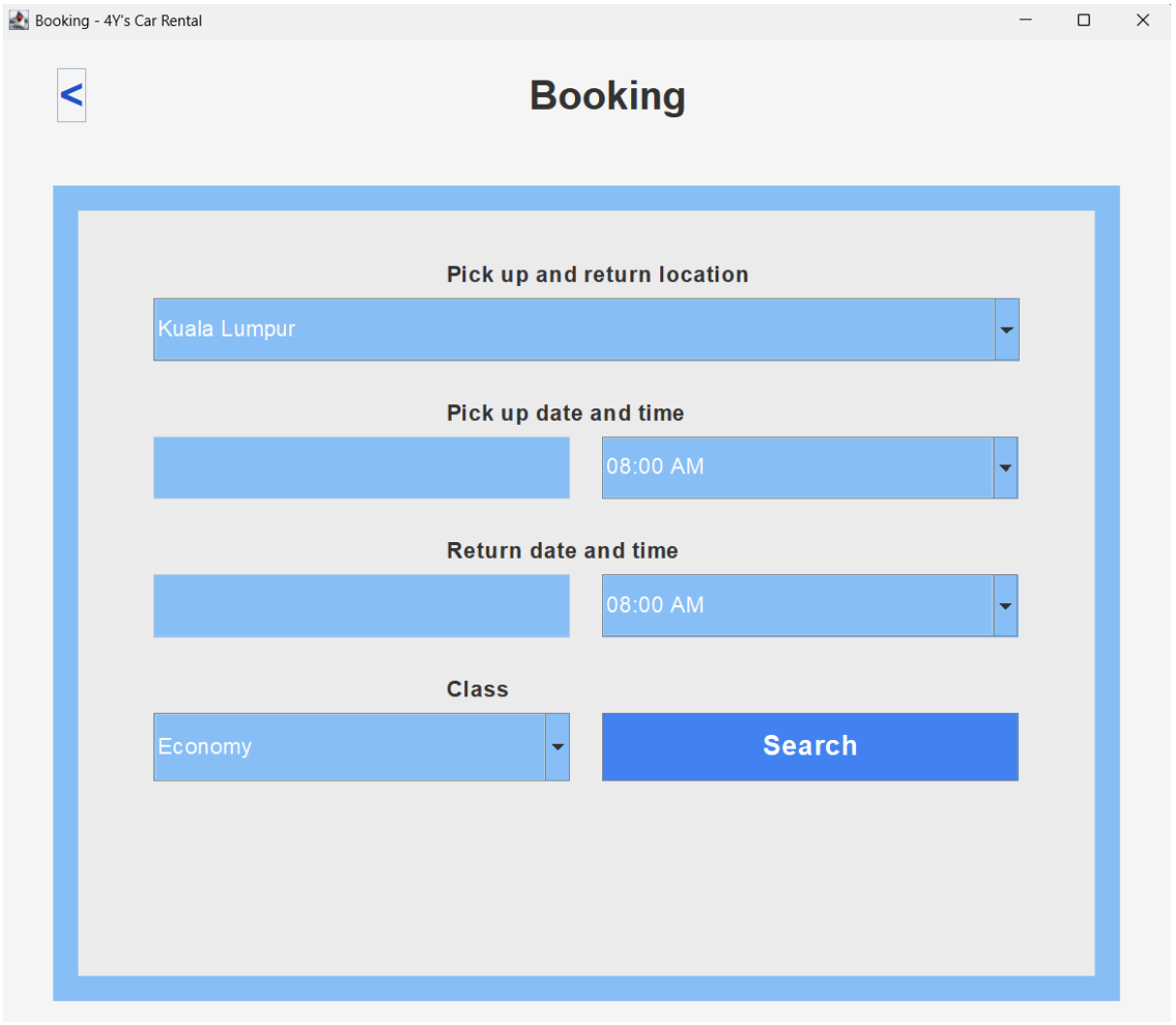
- Allow customers to start a booking for rental cars.

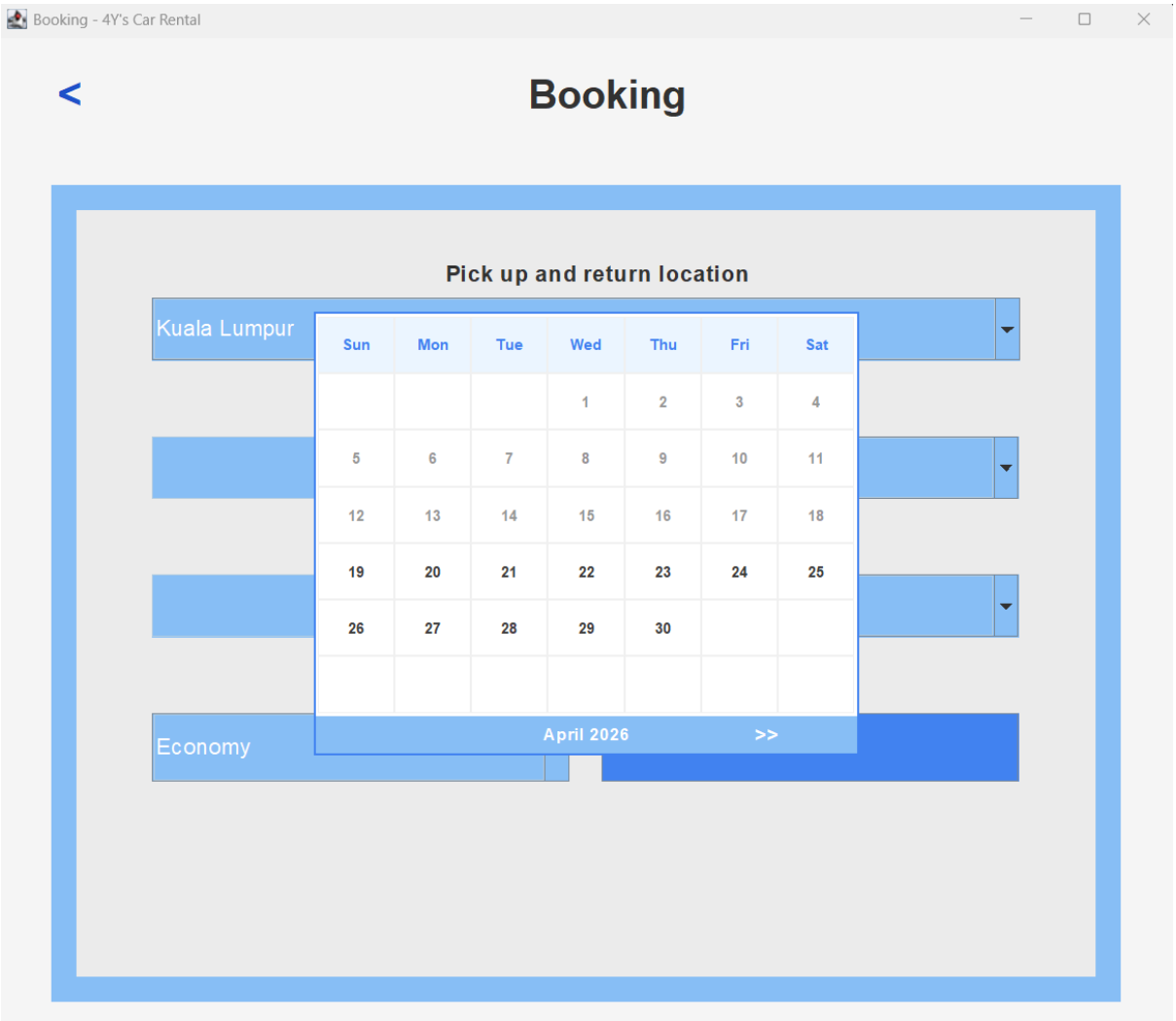
[2] View History

- Allow customers to view their booking history.

4.2.1 Customer Booking

4.2.1.1 Booking Page





In the Booking Page, it will display the pick up and return location, pick up date and time, return date and time and class. Customers can choose the location, pick up date and time they like. When customers choose the pick up date and return date, the system will display a calendar.

Steps:

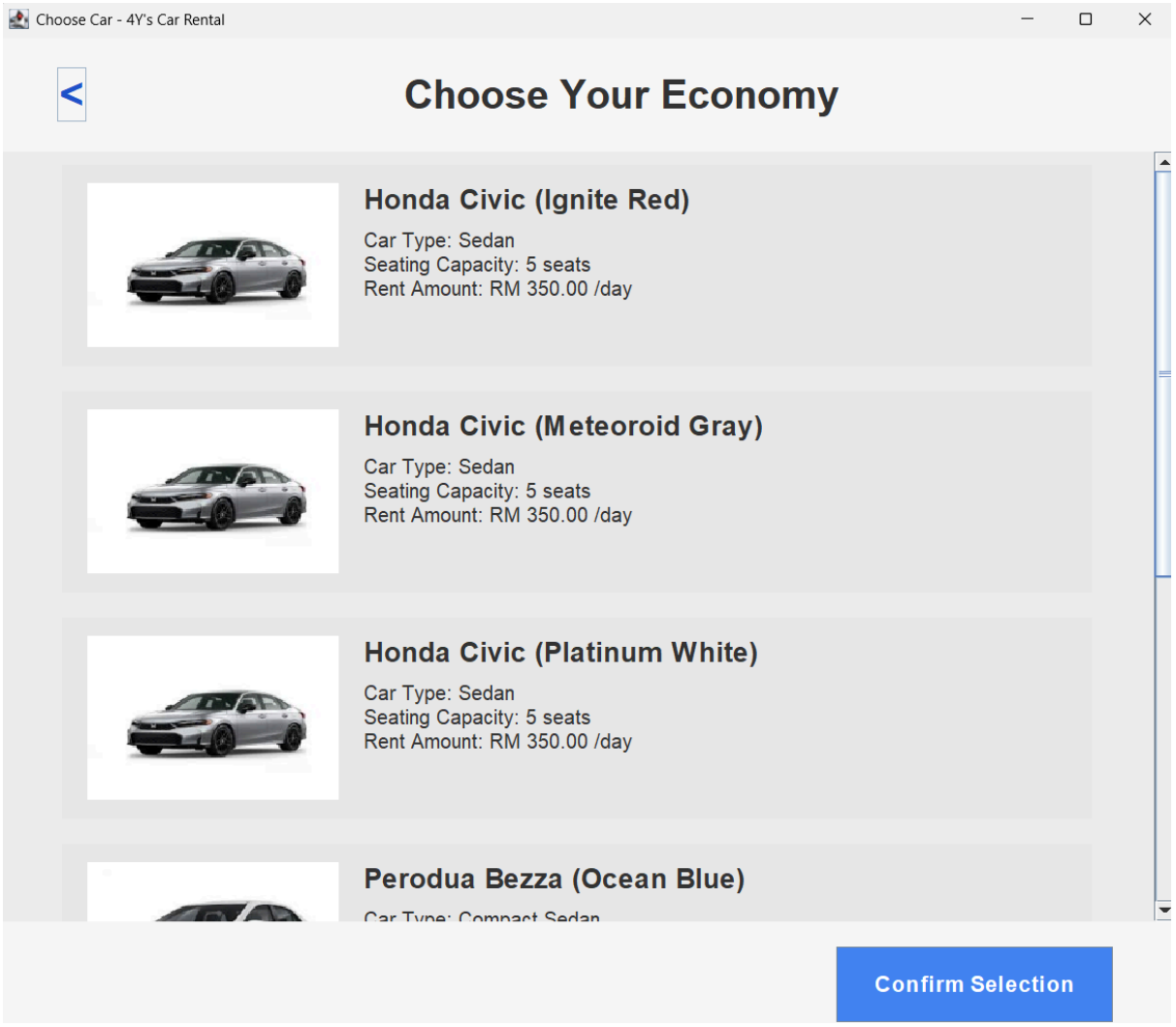
1. Customer need to select a **pick up and return location**.
2. Customer enter a **pick up date** and select the **time**.
3. Customer enter a **return date** and select the **time**.
4. Customer need to select a **class** of the car (Economy, Luxury, RV).
5. After selecting, customer can press the **Search** button to jump to the **Choose Car Page**.

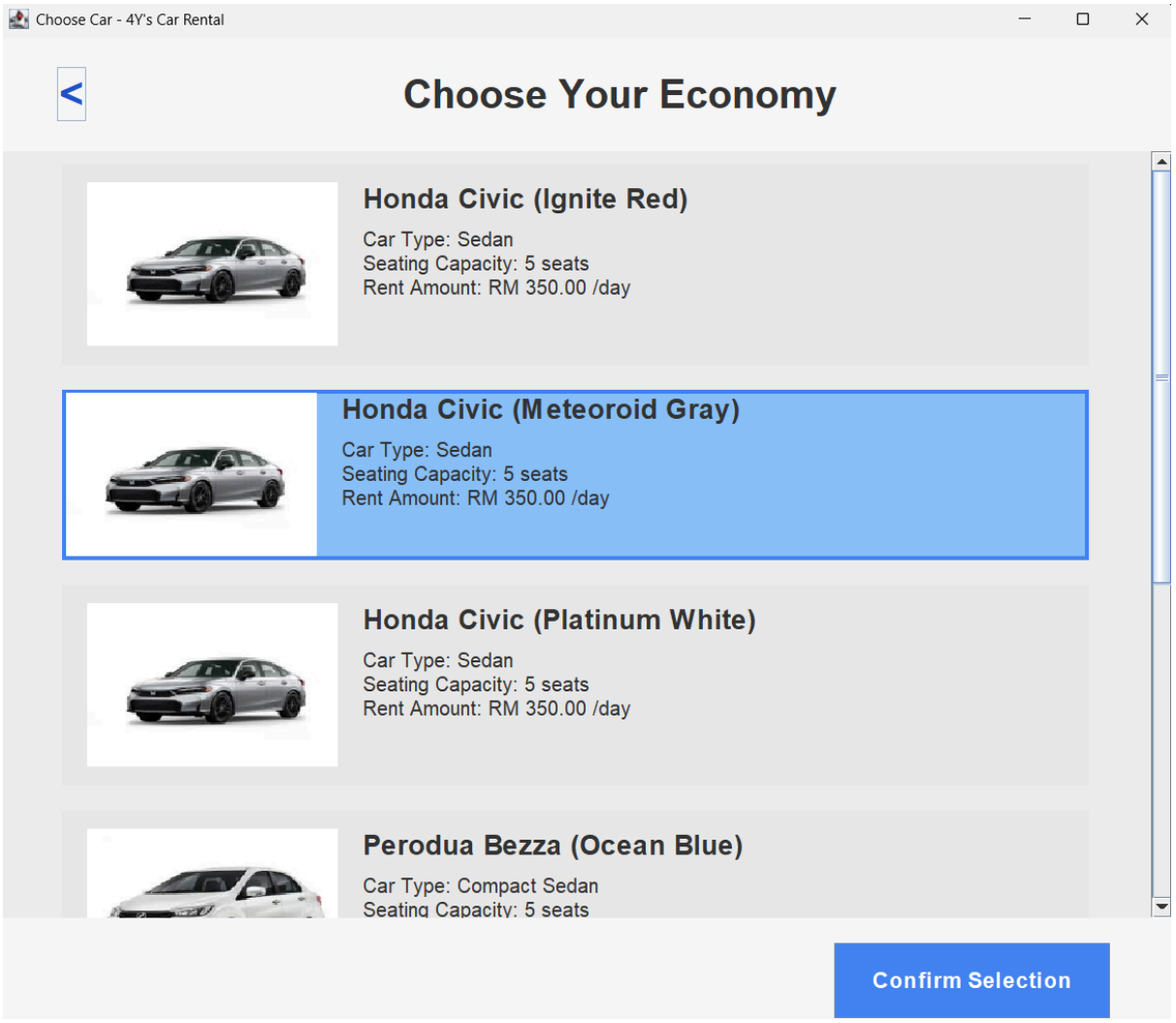
Note:

- If the pick up date and time is earlier than or same with the return date and time, the system will display “Return time must be later than pick up time!”.
- If the pick up date and return date is empty, the system will display “Invalid date! Format: yyyy-MM-dd”.

Customer can also click the **Back** button in the upper left corner to return to the **Customer Main Page**.

4.2.1.2 Choose Car Page





// after select a car

In the Choose Car Page, it will display the car info (car image, car name, car type, seating capacity and rental amount). Customers can choose the car they want to rent.

Steps:

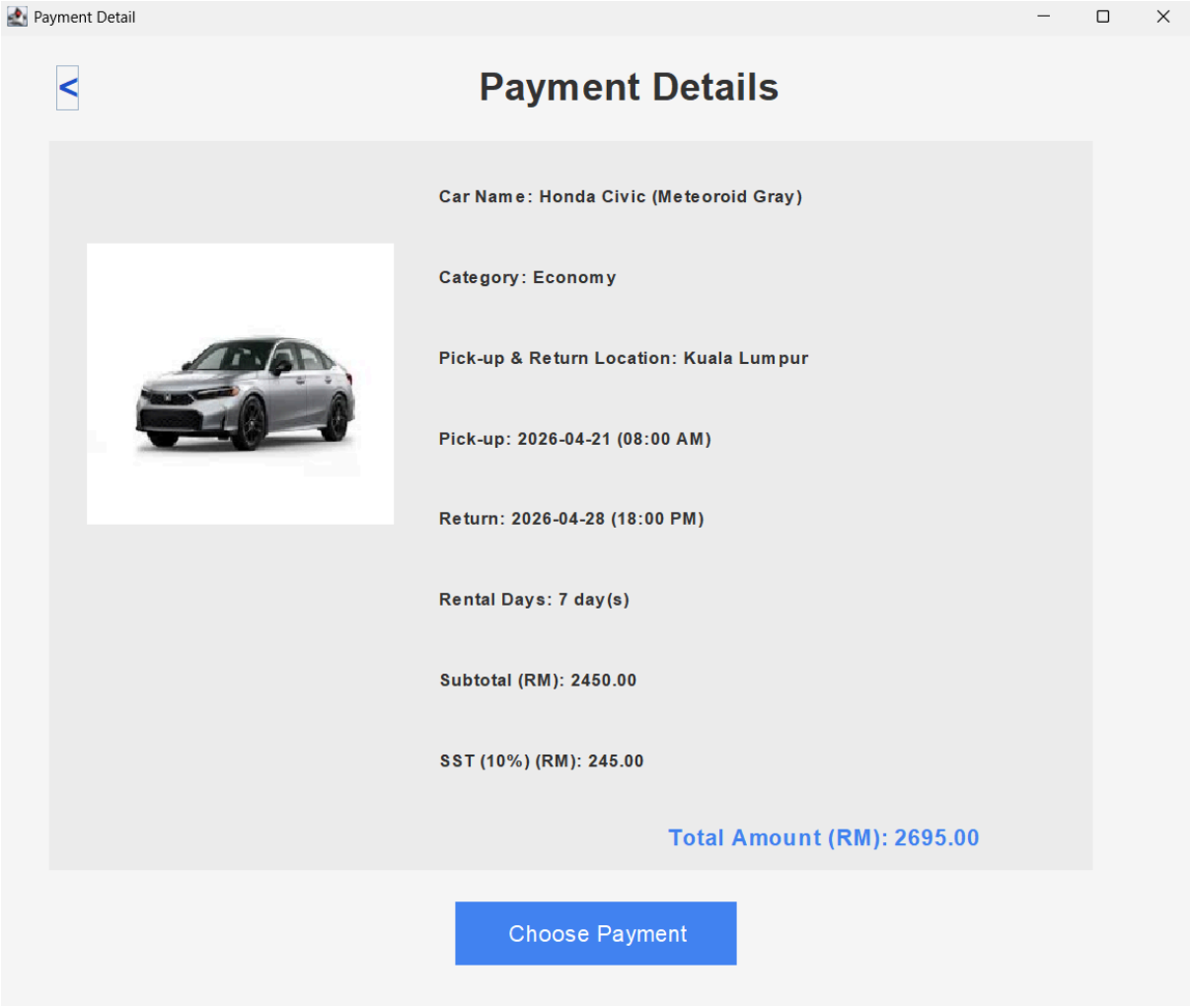
1. Customer select a car.
2. After selecting, customer can press the **Choose** button to jump to the **Payment Details Page**.

Note:

- If the customer does not select a car but clicks the Choose button directly, the system will display “Please select a car before proceeding!”.

Customer can also click the **Back** button in the upper left corner to return to the **Booking Page**.

4.2.1.3 Payment Page

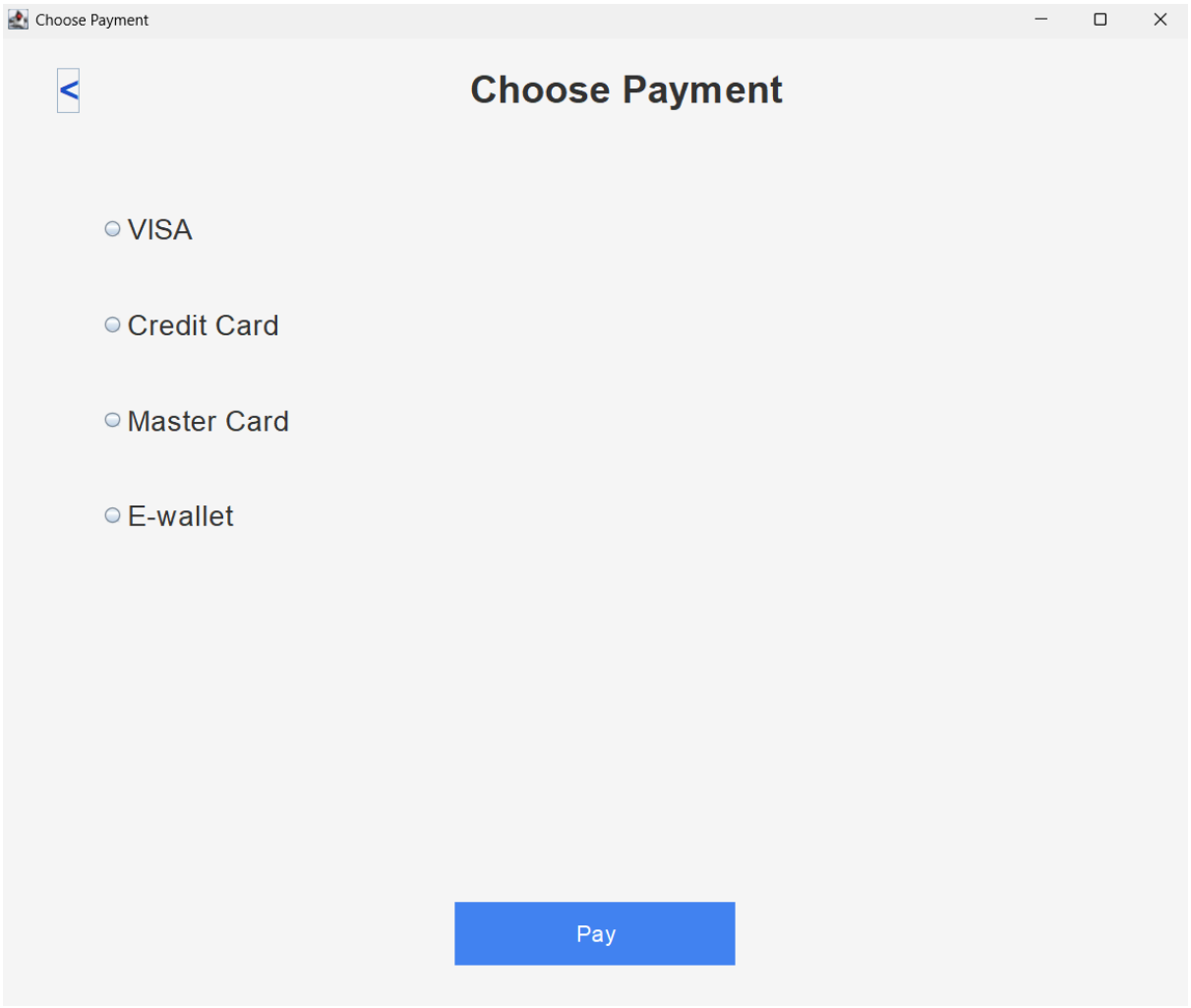


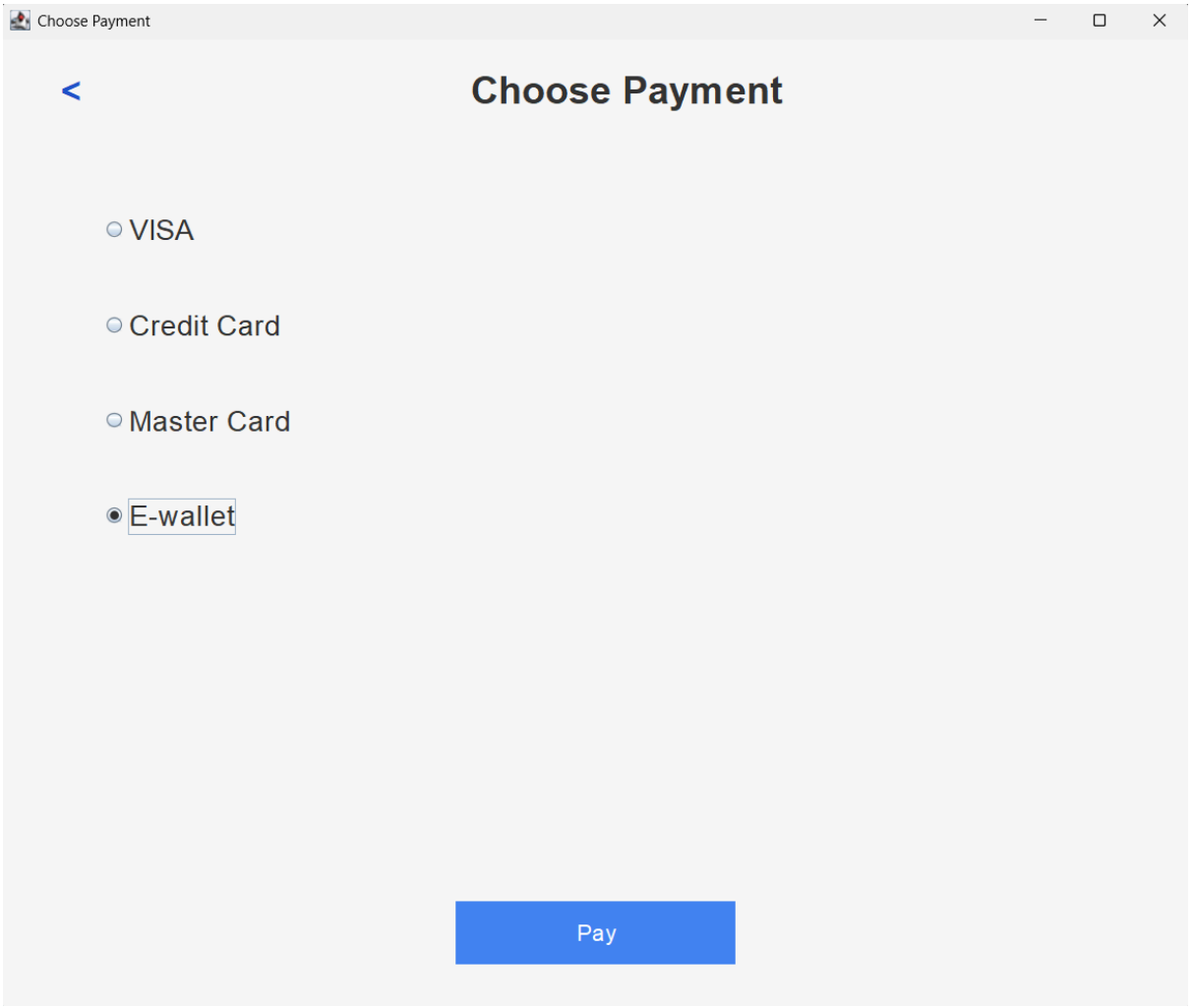
In the Payment Details Page, it will display the selected car info and the booking info. It will also display the rental days (calculated by the pick up date and the return date) and the total amount (calculated by subtotal and SST).

Steps:

1. Customers need to check if the booking details are what they selected.
2. If correct, the customer can press the **Choose Payment** button to jump to the **Choose Payment Page**; If incorrect, the customer can click the **Back** button in the upper left corner to return to the **Choose Car Page** and make the selection again.

4.2.1.4 Choose Payment Page





// *after select a payment method*

In the Choose Payment Page, customer can choose a payment method to make a payment. There are four payment methods, which are VISA, Credit Card, Master Card and E-wallet.

Steps:

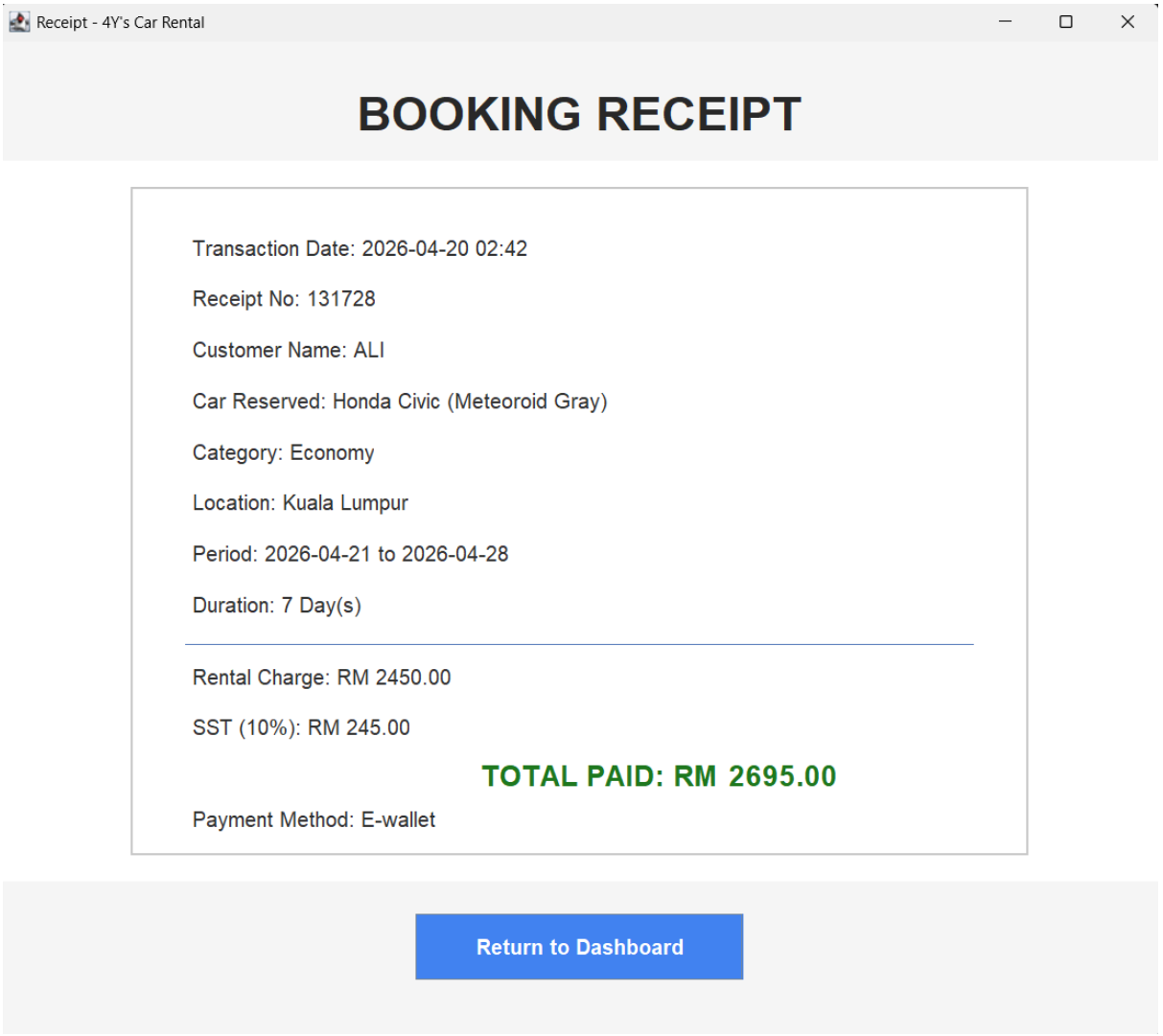
1. Customer select a payment method.
2. After selecting, customer can press the **Pay** button to jump to the **Receipt Page**.

Note:

- If the customer does not select a payment method but clicks the Pay button directly, the system will display “Please select a payment method!”.

Customer can also click the **Back** button in the upper left corner to return to the Payment Details Page.

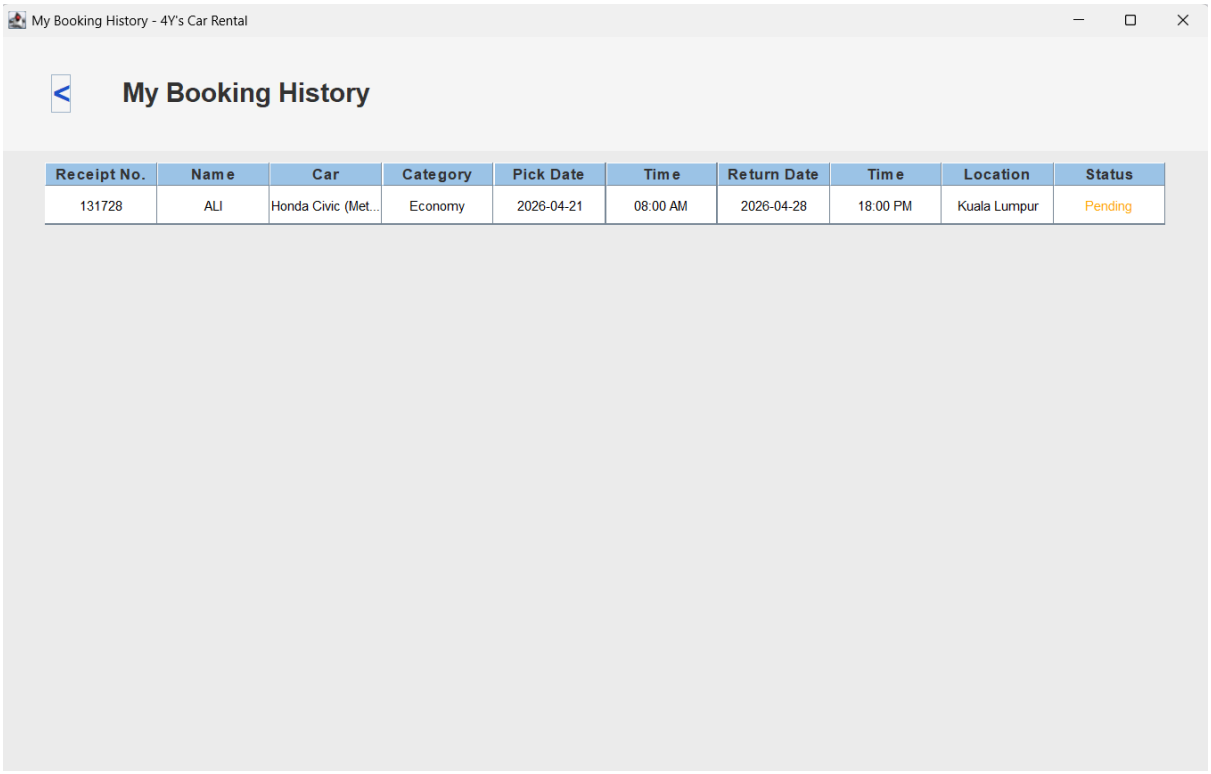
4.2.1.5 Receipt Page



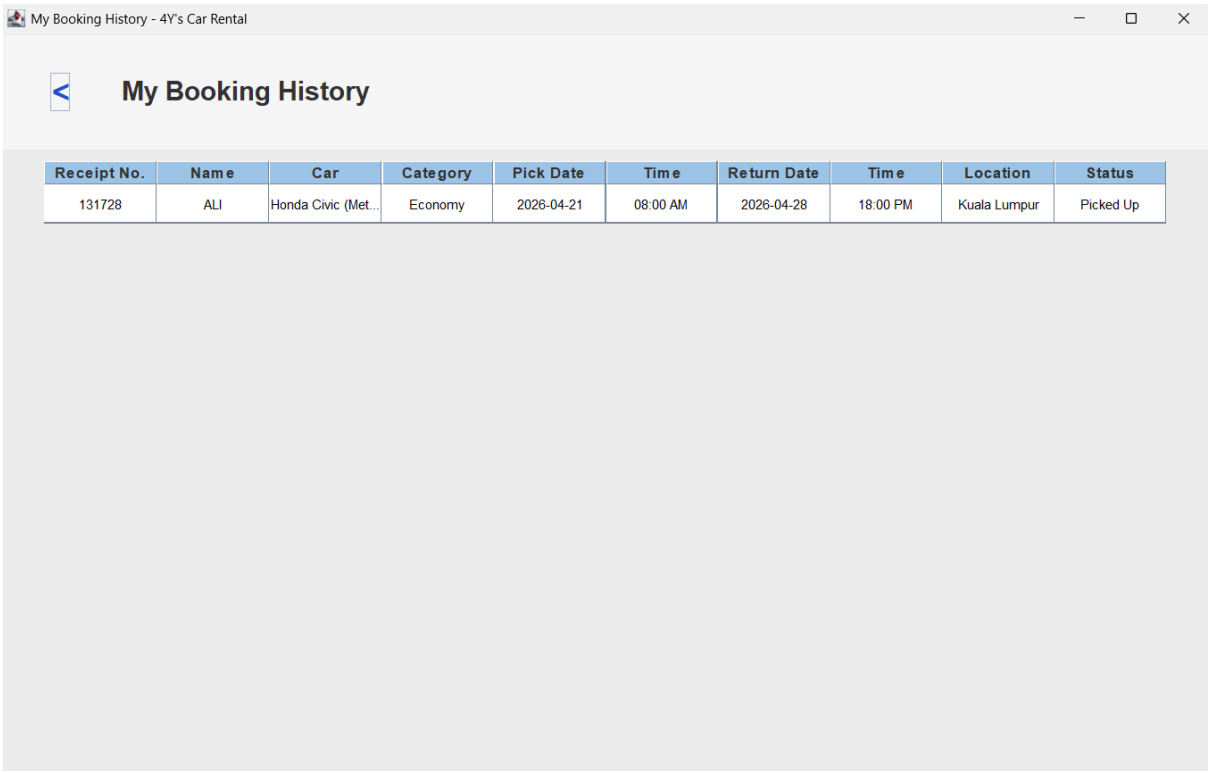
Receipt Page will display the transaction date, receipt no, customer name, selected car info and booking info. It also displays the selected payment method and status.

- Customer can press the **Return to Main Page** button to jump to the **Customer Main Page**.

4.2.2 Customer View History

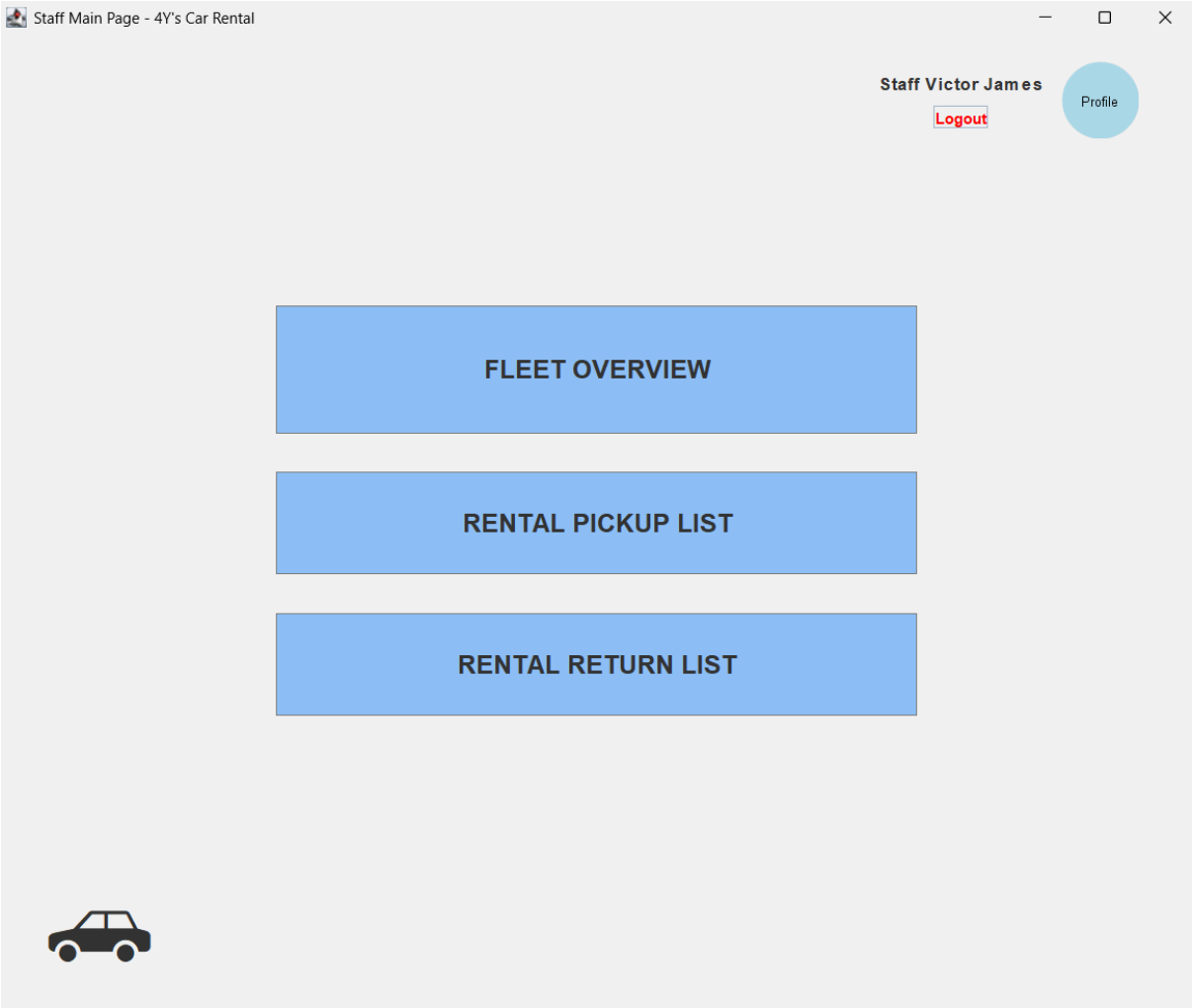


In the Booking History Page, it will display the receipt no, customer name, customer's booking details, selected car info and status, customer can check their booking history here. The status will show the rental car status (Returned, Pending, Late Returned).



After staff change the status, the status will display "Picked Up" in the Status row.

4.3 Staff Management



The Staff Main Page includes three options, which is fleet overview, rental order list and rental return list.

The menu option include:

[1] Fleet Overview

- Allow staff to manage every vehicle and report a problem about the vehicle.

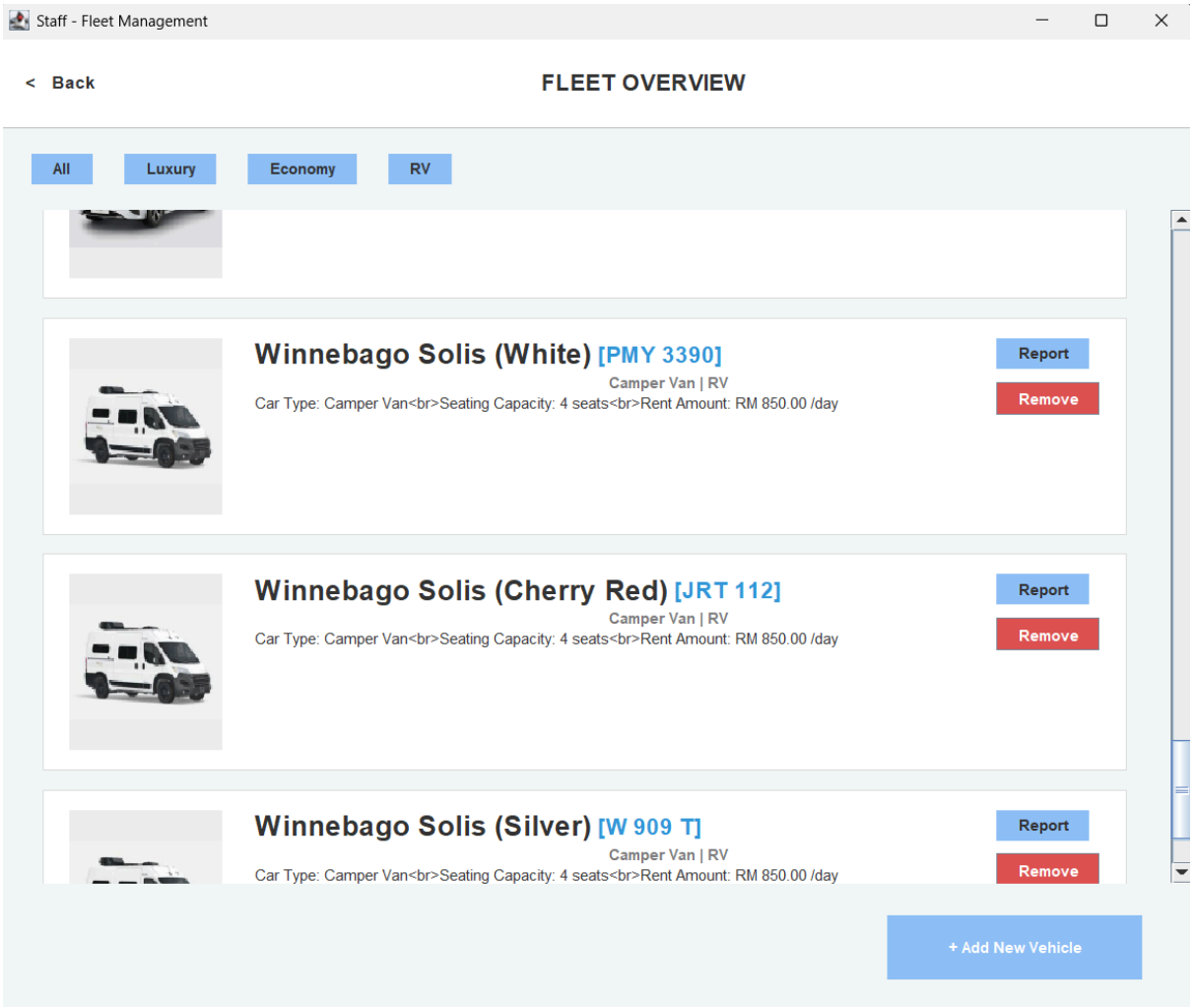
[2] Rental Order List

- Allow staff to manage the customer booking orders.

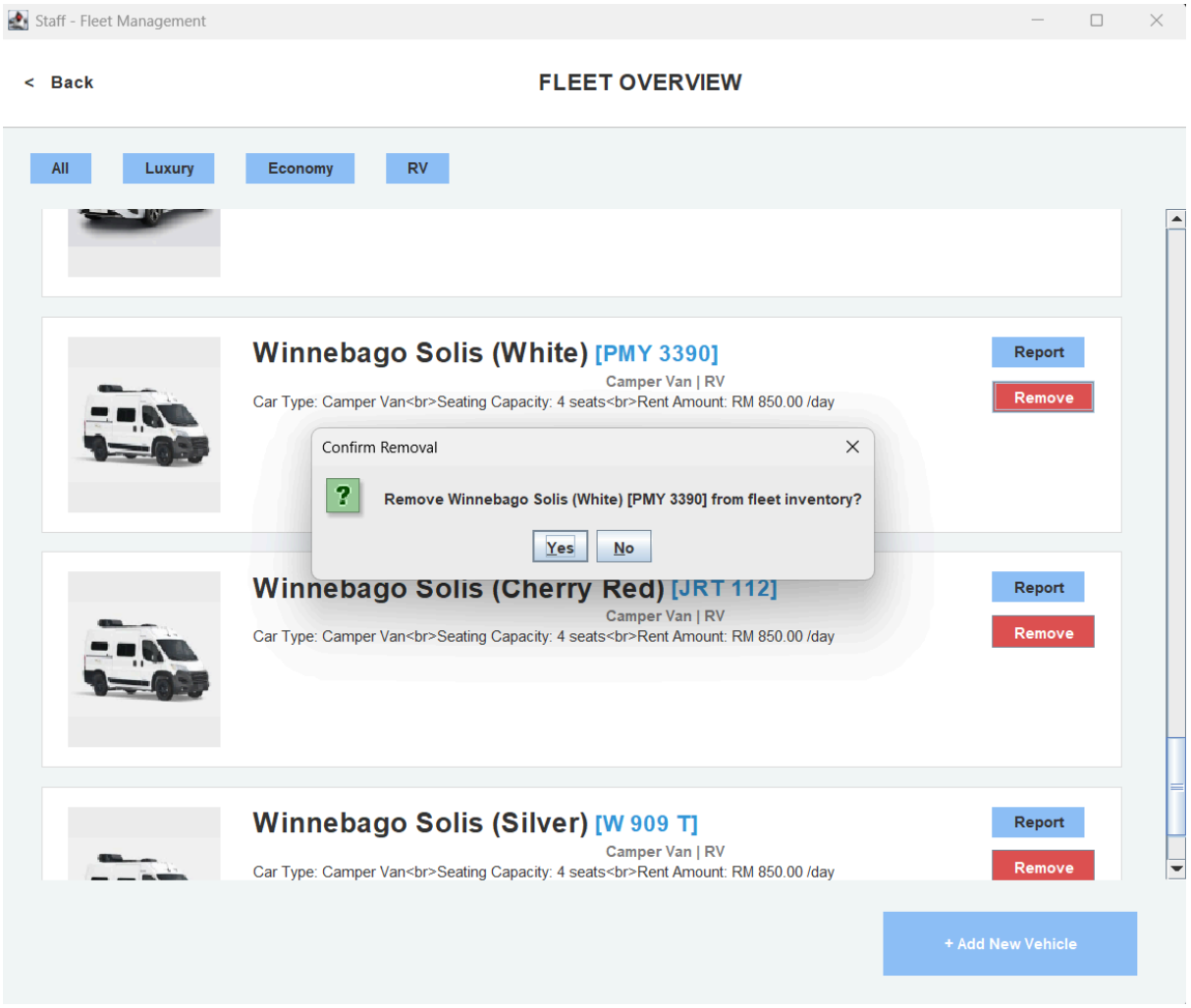
[3] Rental Return List

- Allow staff to manage the customer booking returns.

4.3.1 Fleet Overview



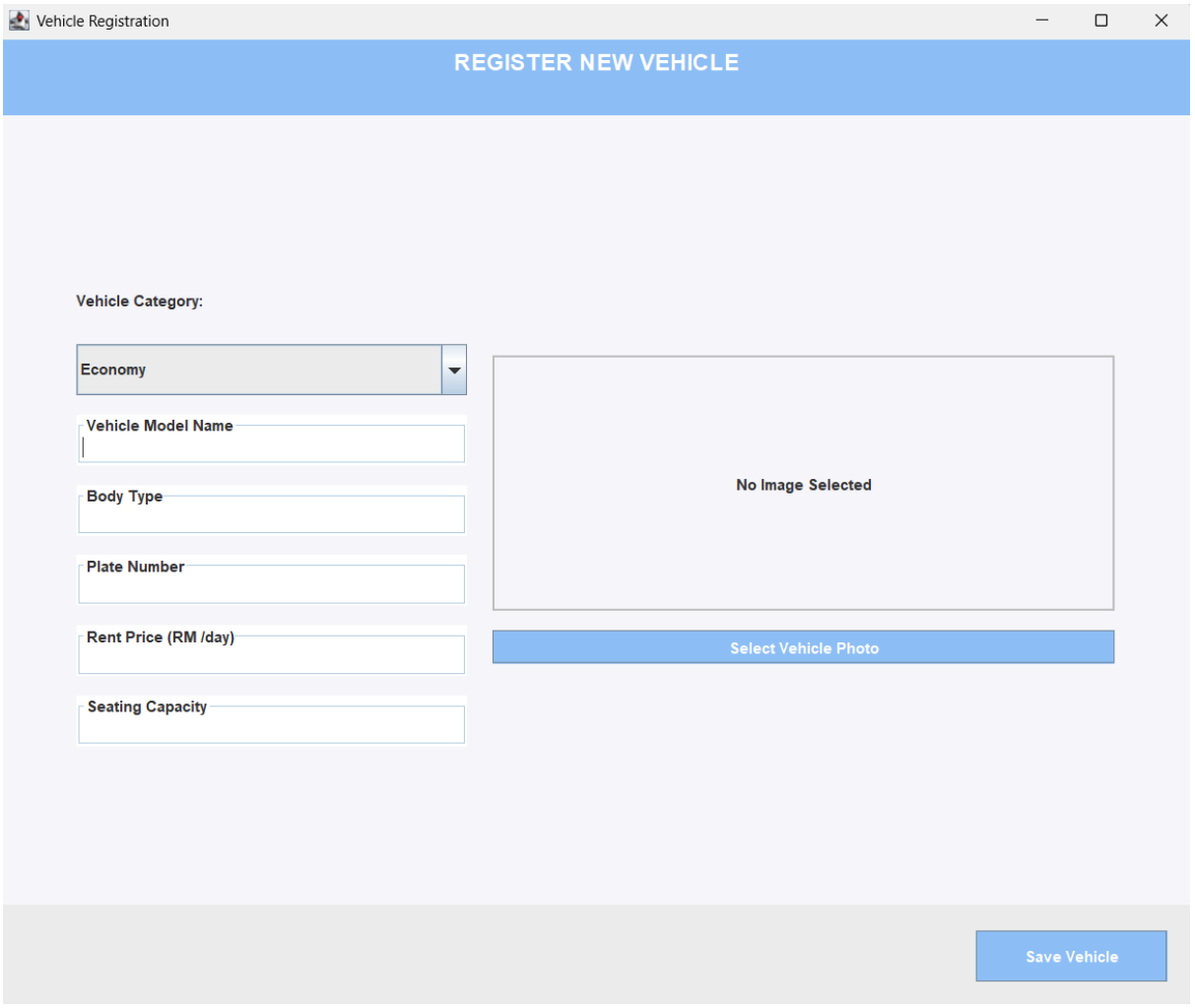
In the Fleet Overview Page, staff can add a new car with adding car image, car name, car plate number and other details.



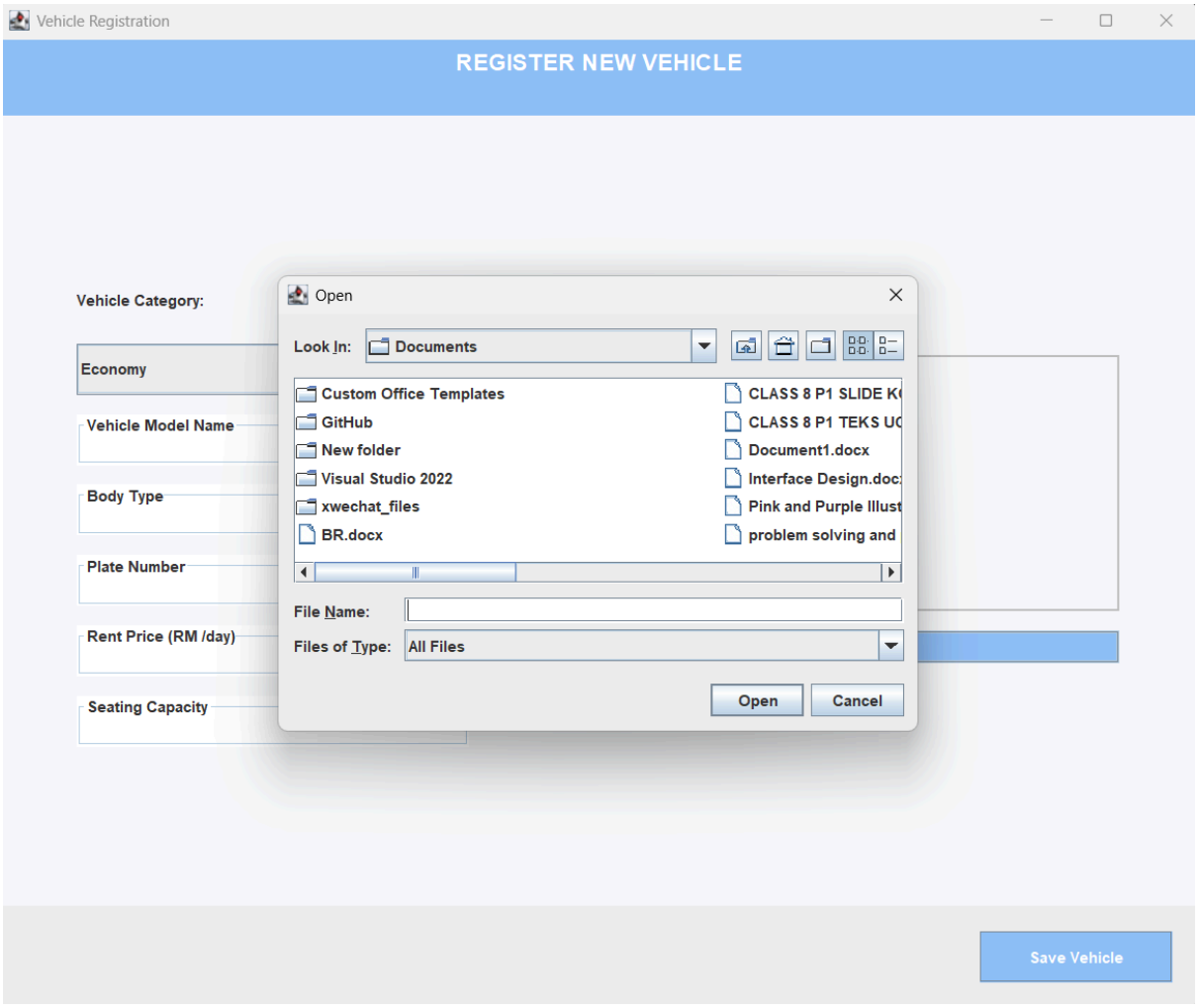
If staff want to remove a car, the system will display a confirmation message to confirm staff action.

Staff can also click the **Back** button in the upper left corner to return to the **Staff Main Page**.

4.3.1.1 Add New Vehicle Page



In this page, staff can add a new vehicle by entering the new car information. Staff need to click the **Select Vehicle Photo button** to add a car photo. After entering all information and photo, staff need to press the Save Vehicle button to save it.



// After click Select Vehicle Photo button

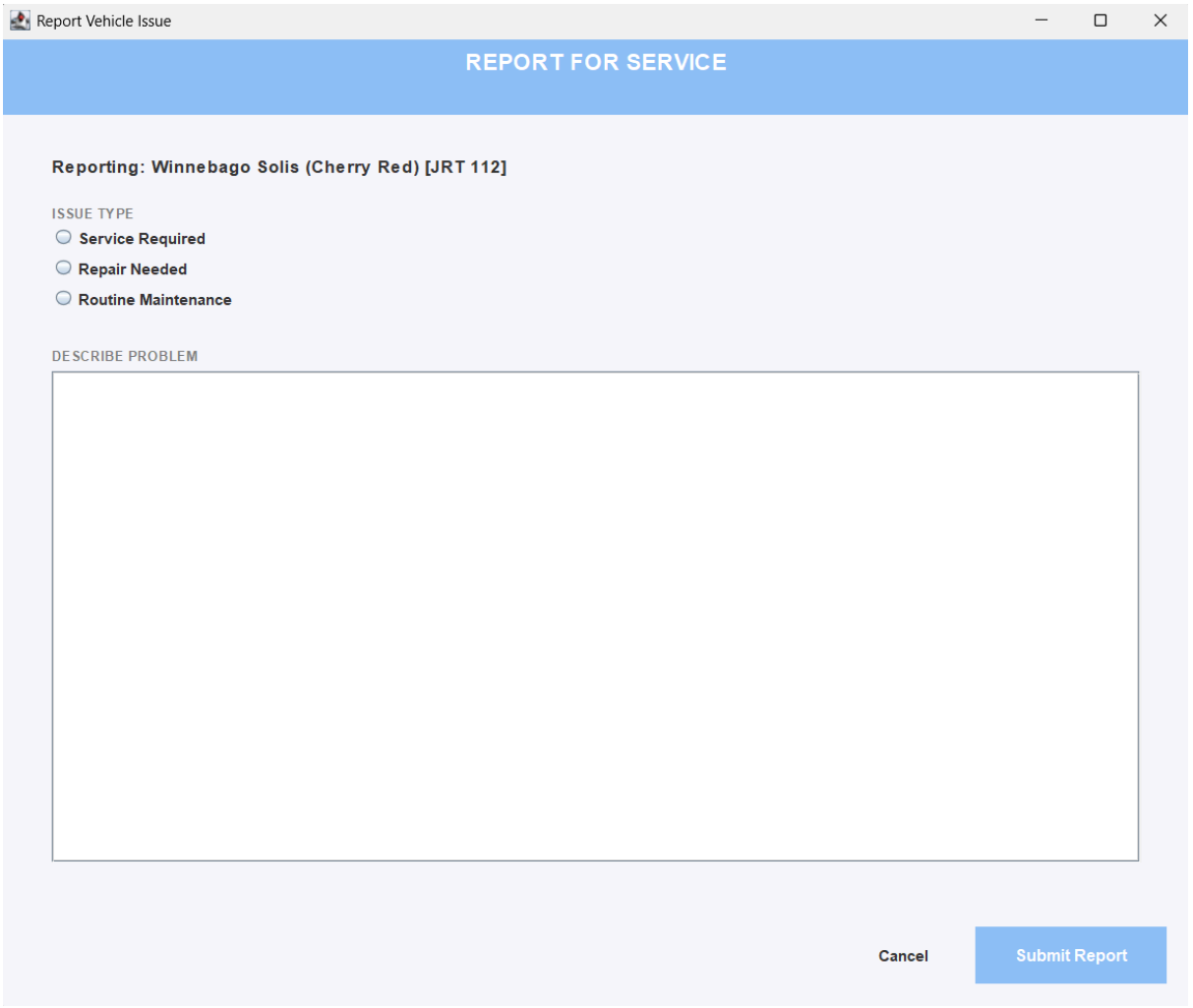
Steps:

1. Staff select a vehicle category.
2. Staff enter new car information (Vehicle Model Name, Body Type, Plate Number, Rent Price, Seating Capacity).
3. Staff select a car photo.
4. After entering, staff can press the **Save Vehicle** button to save the new car information.

Note:

- If the car information is empty, the system will display “Model Name and Image are Required!”

4.3.1.2 Report for Service Page



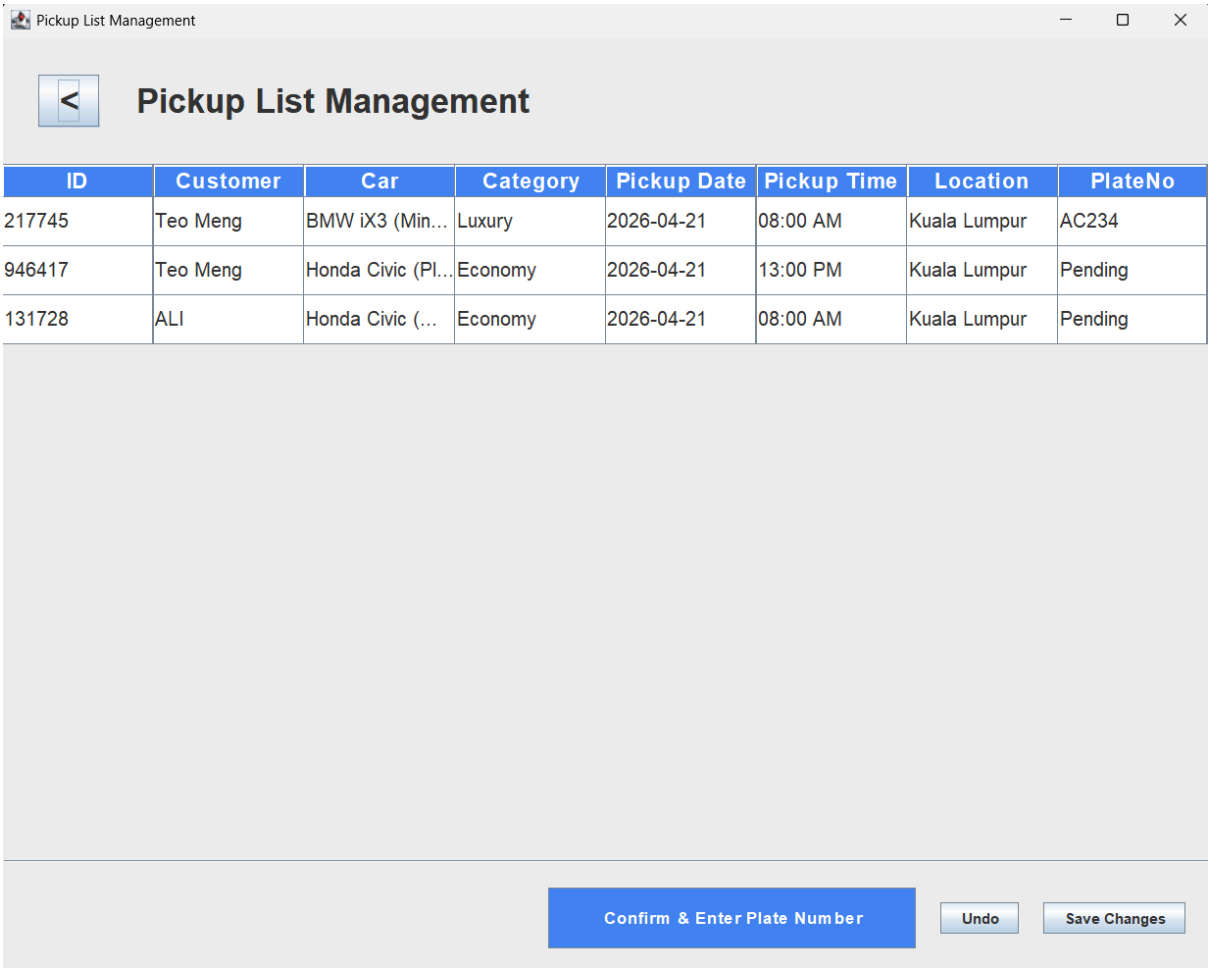
Besides, staff can also report a car for service. In the Report Page, staff can choose a reason and describe the problem.

Steps:

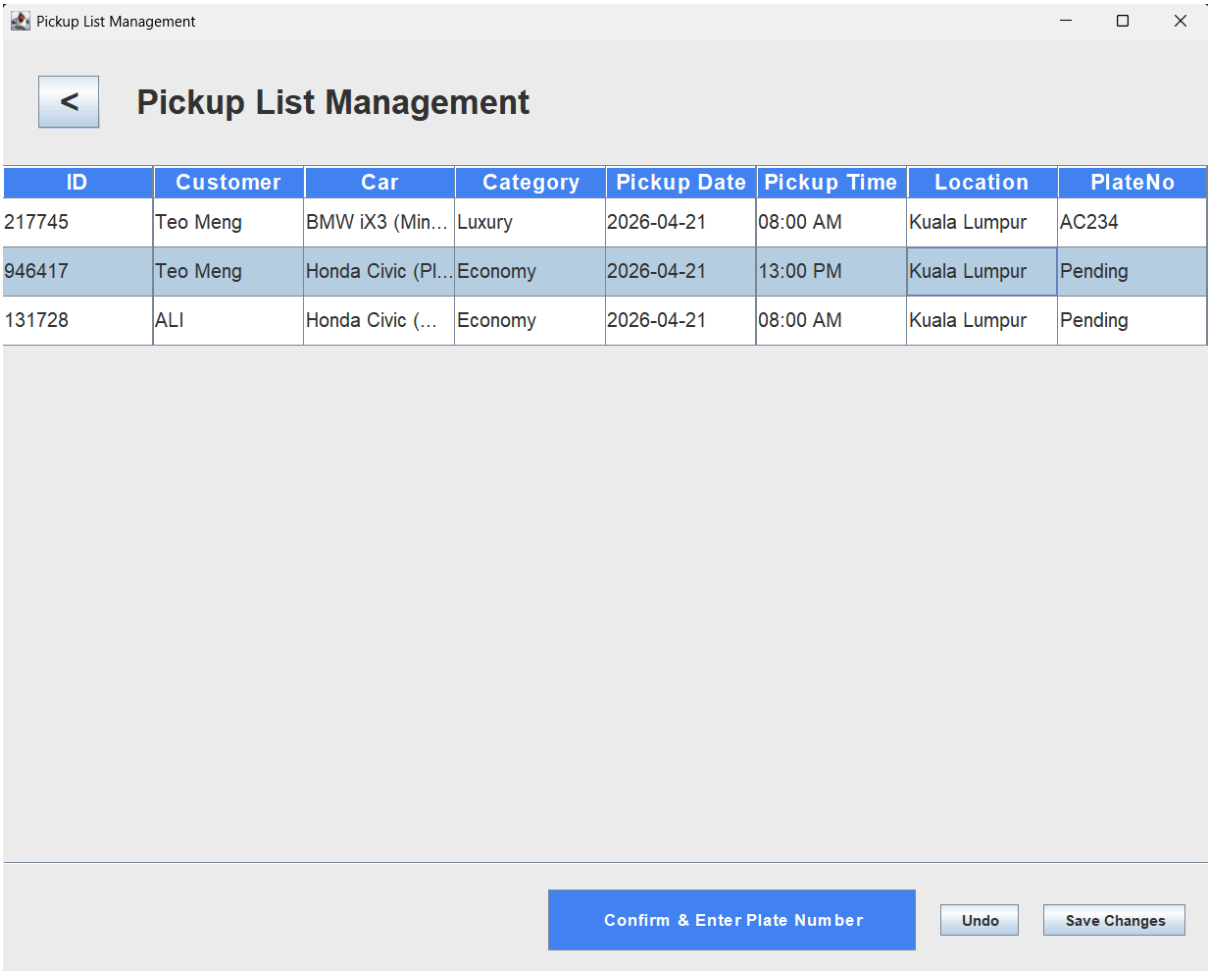
1. Staff choose an issue type.
2. Staff describe problems in the blank space.
3. After entering the problems, staff can press the **Submit Report** button

Staff can press the cancel button to jump to the **Fleet Overview Page.**

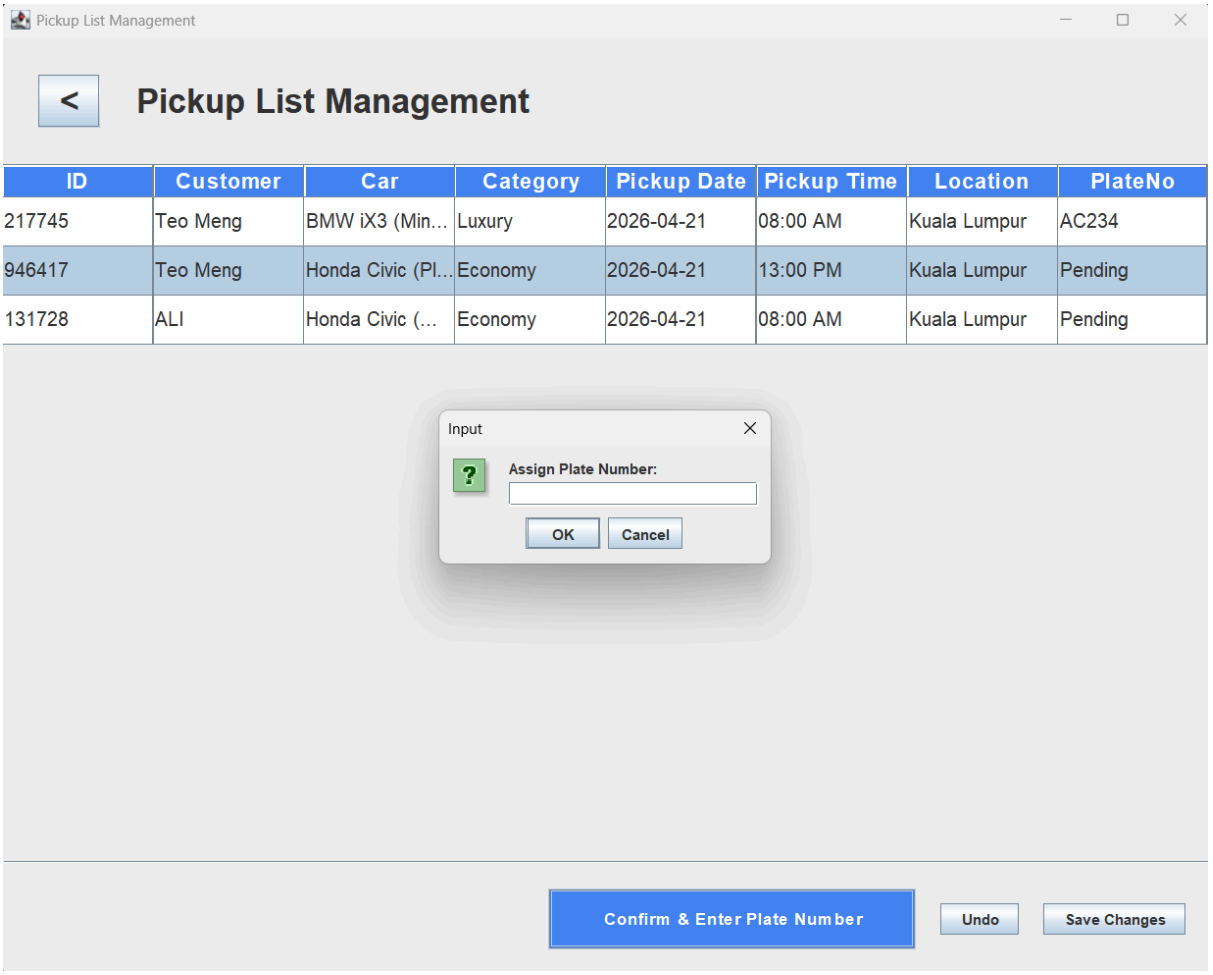
4.3.2 Rental Pickup List



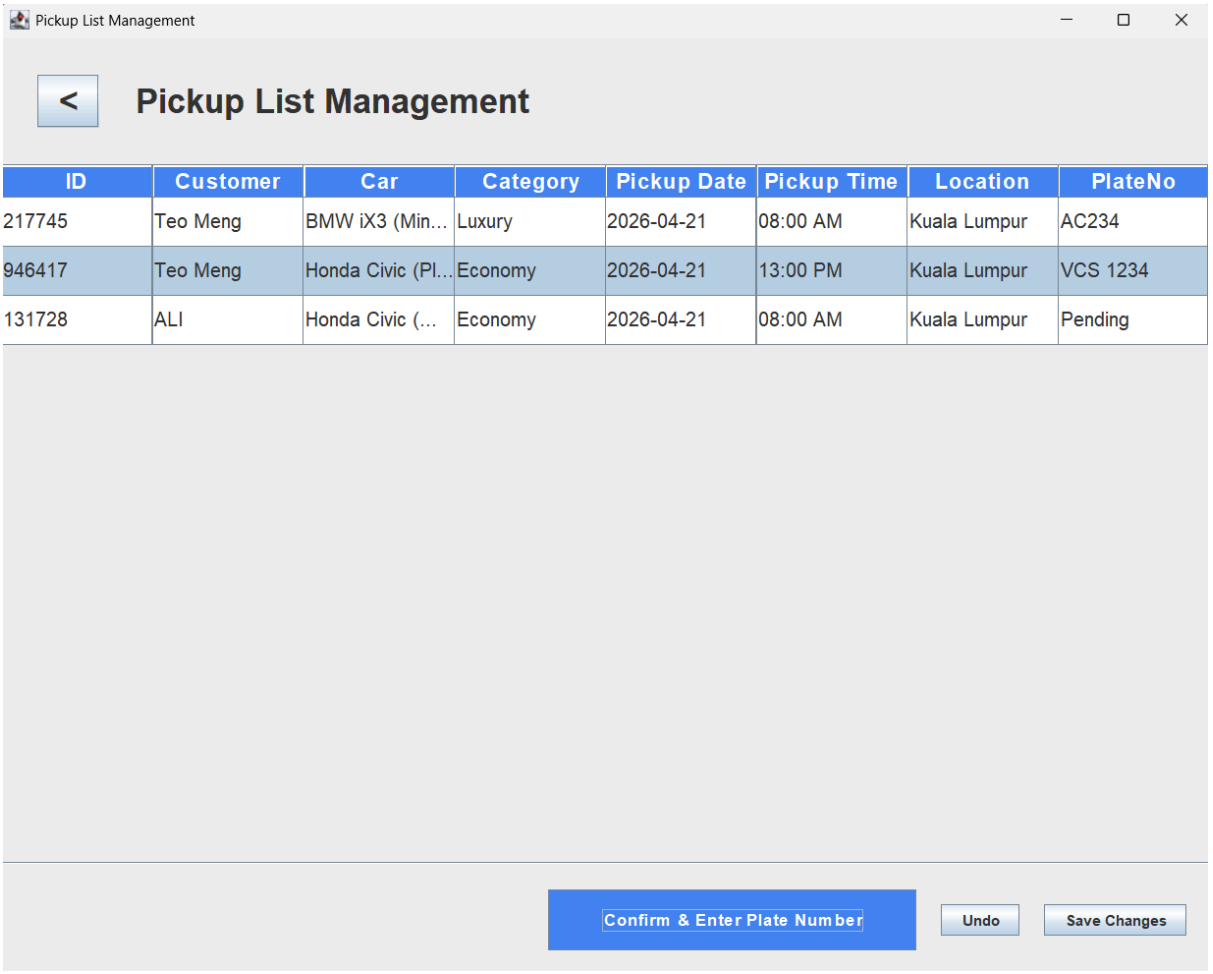
In the Rental Pickup List Page, staff can view the booking history from the customer. It will display receipt no (ID), customer name, car information, booking info and plate number. The Plate Number is default “Pending”, staff be responsible to change the PlateNo. This page allows staff to monitor and manage all current and past orders.



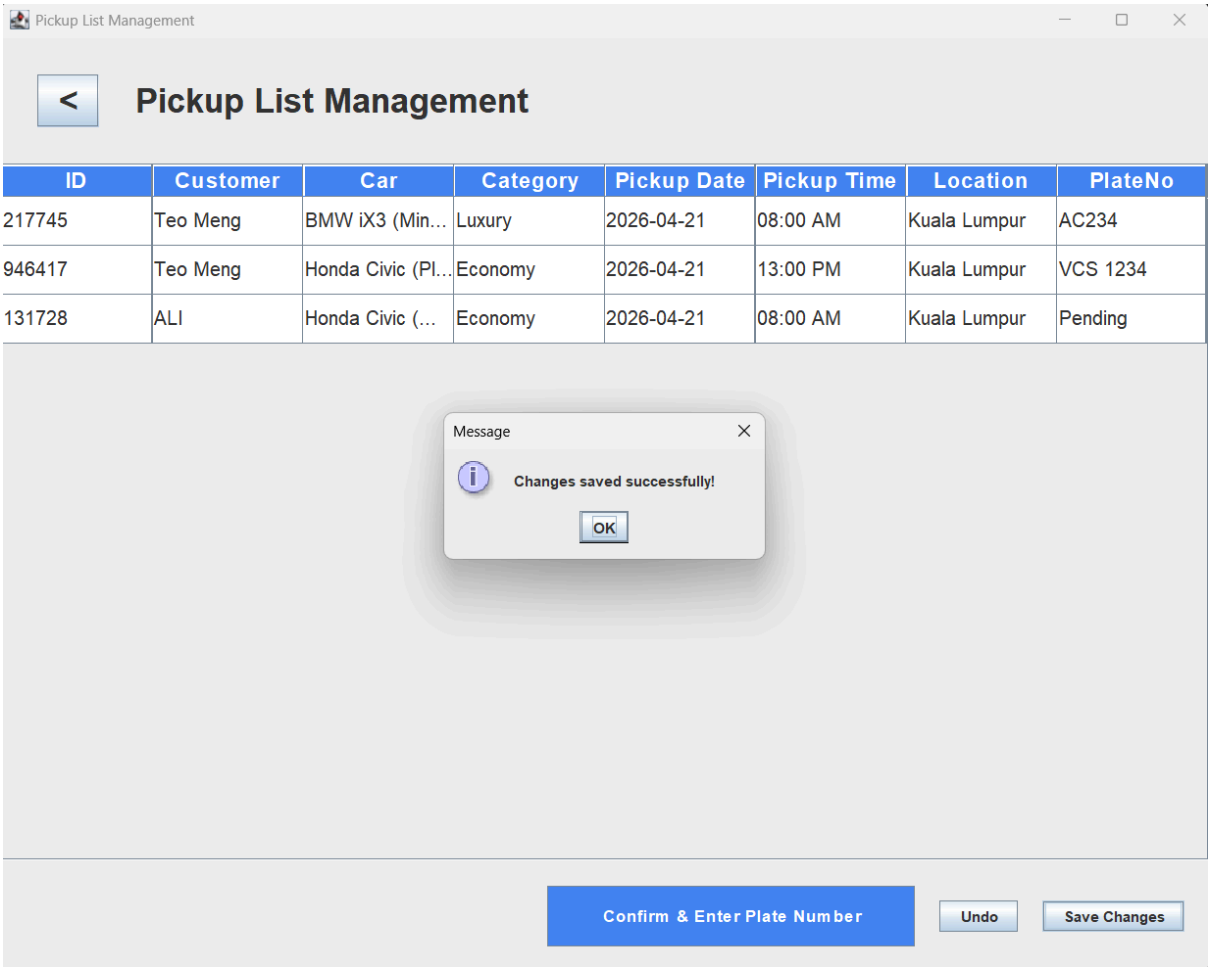
// After select a row



// After press the Confirm & Enter Plate Number button



// After press the OK button



// System display message

Steps:

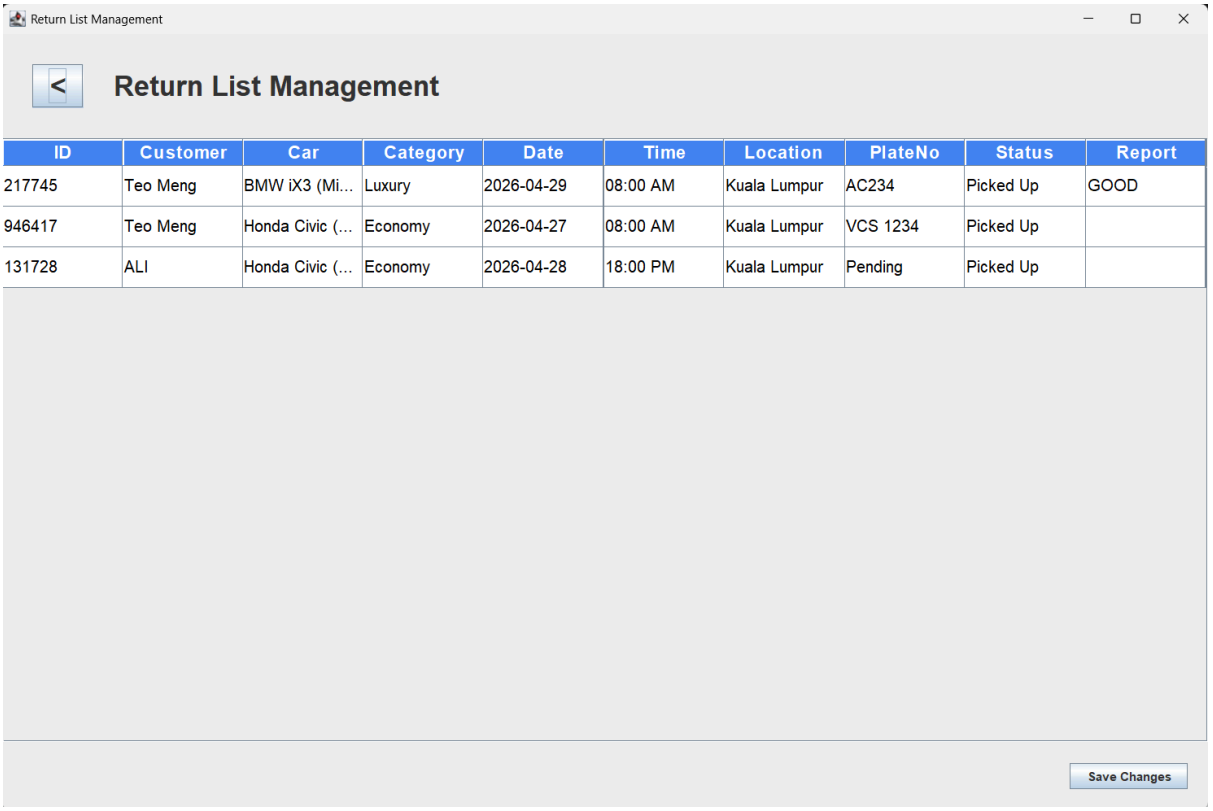
1. Staff select a row.
2. Staff press the Confirm & Enter Plate Number button.
3. Staff assign plate number.
4. Staff press the OK button.
5. The system will display “Changes saved successfully!”
6. After entering, the PlateNo row will change the “Pending” to the plate number.

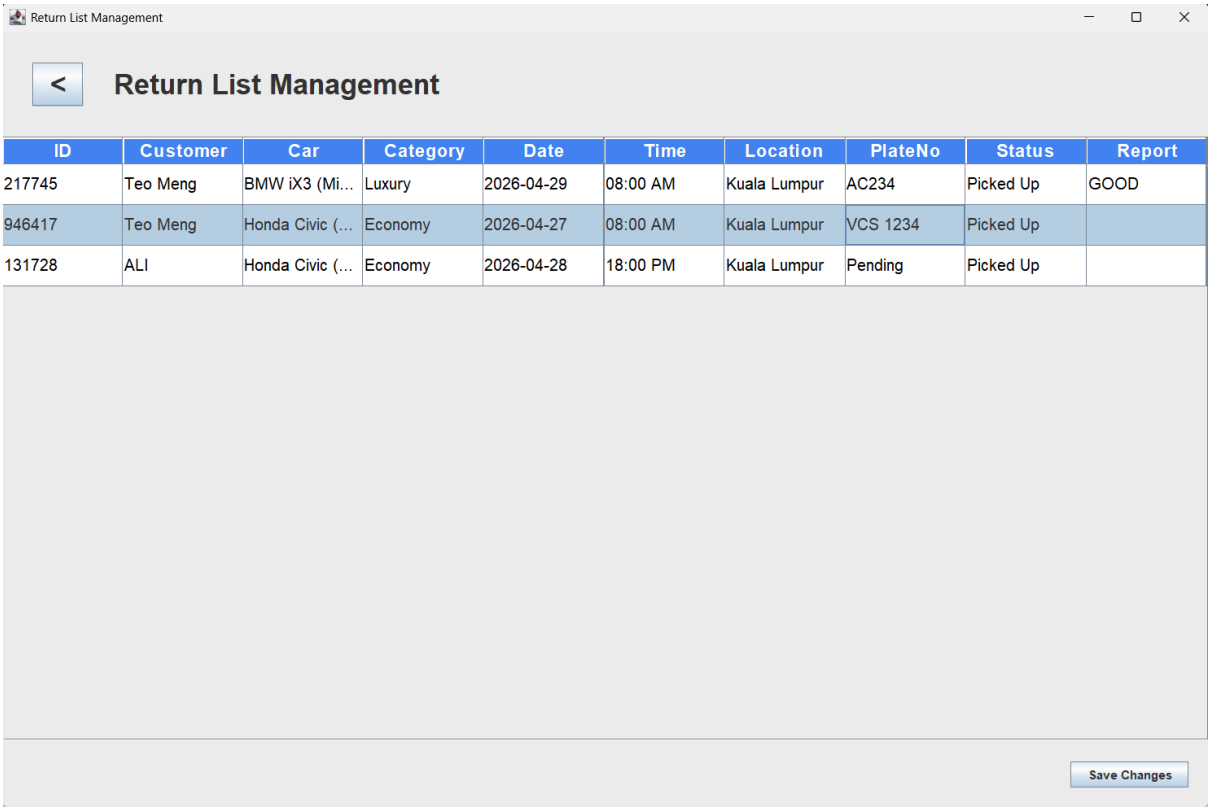
Note:

- If no row is selected, the system will display “Please select a row first!”.

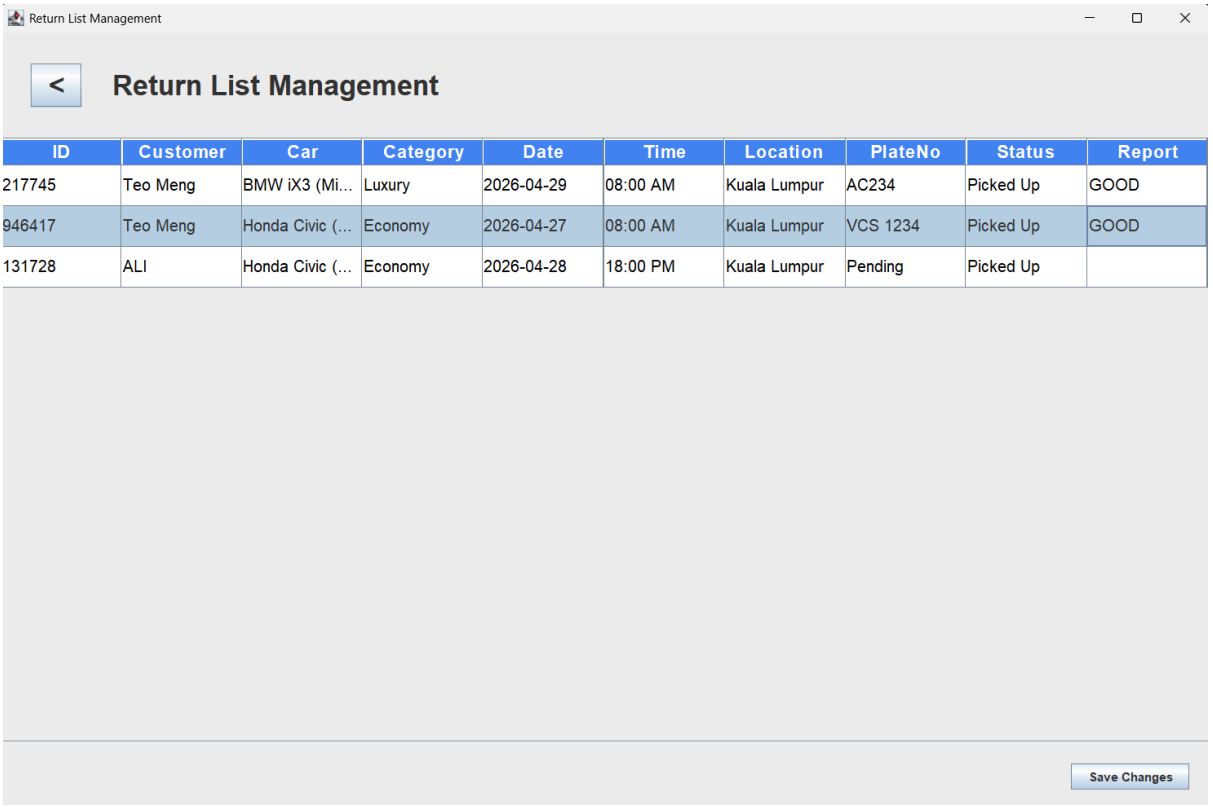
Staff can also click the **Back** button in the upper left corner to return to the **Staff Main Page**.

4.3.3 Rental Return List

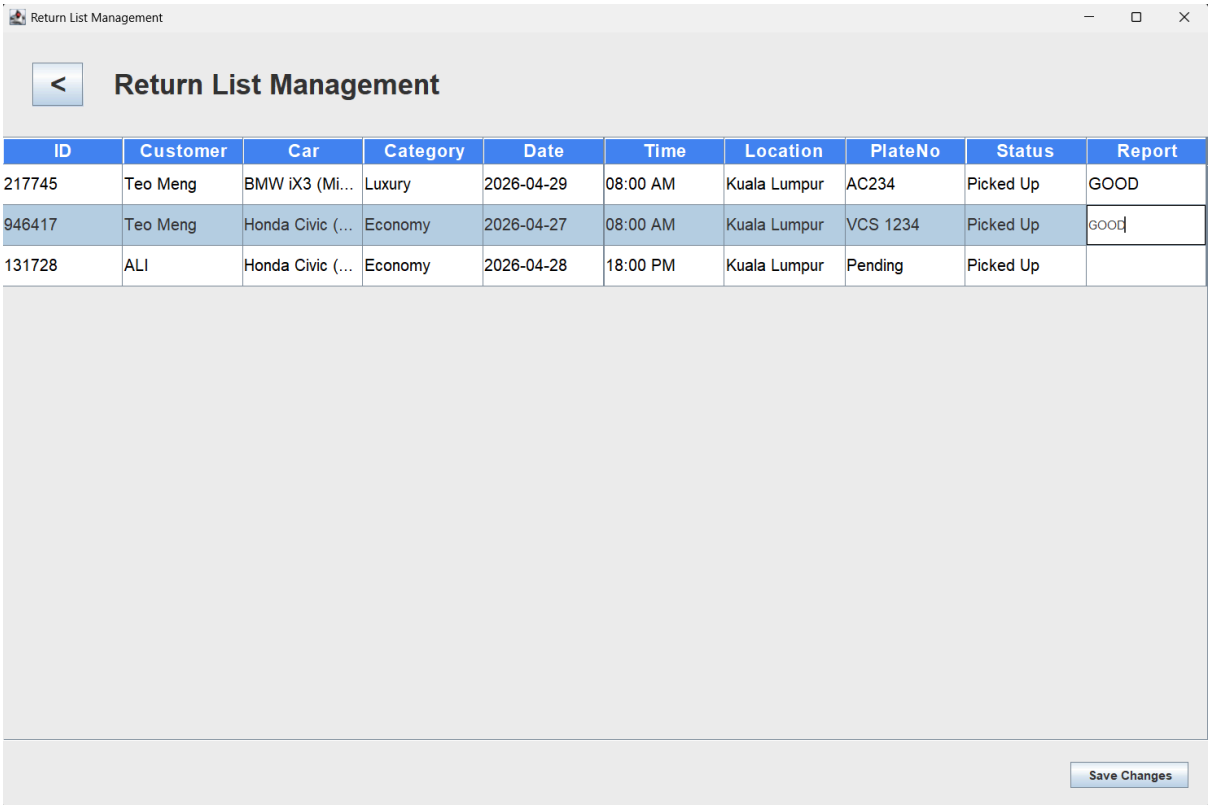




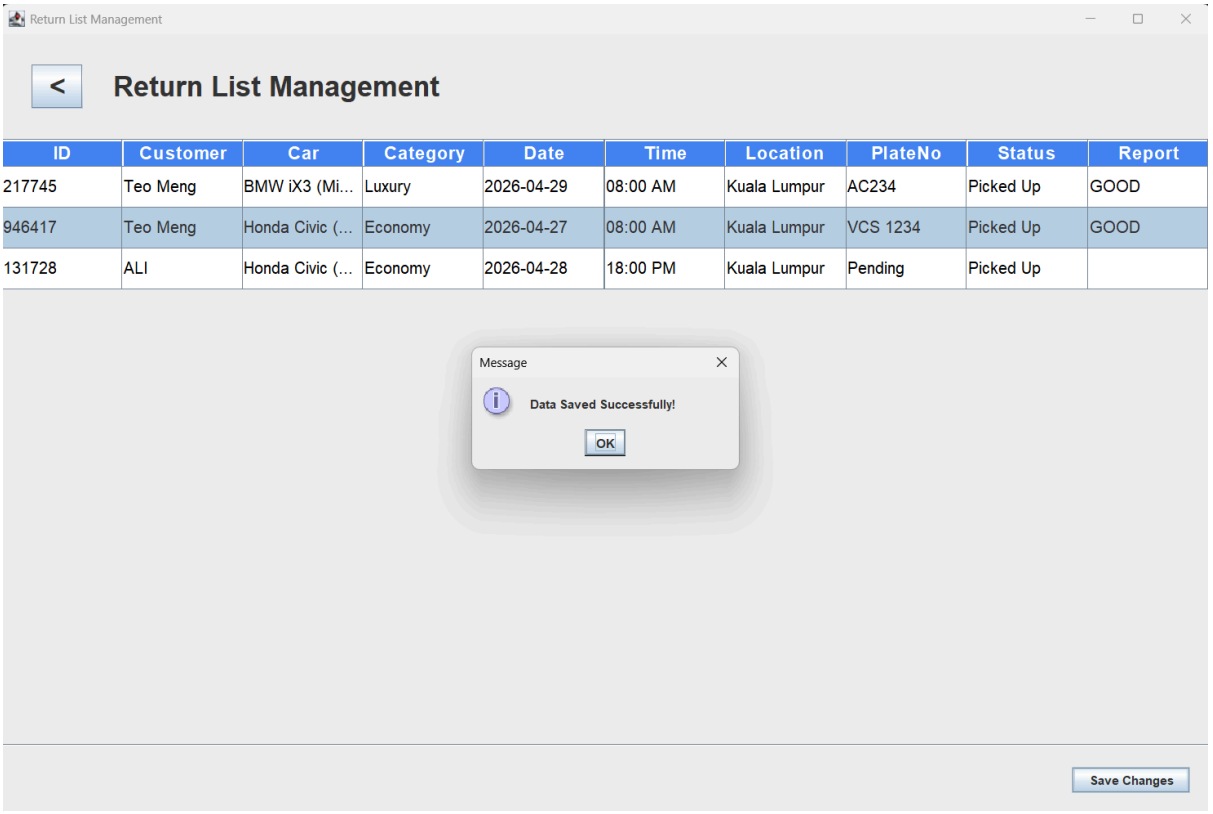
// After select a row

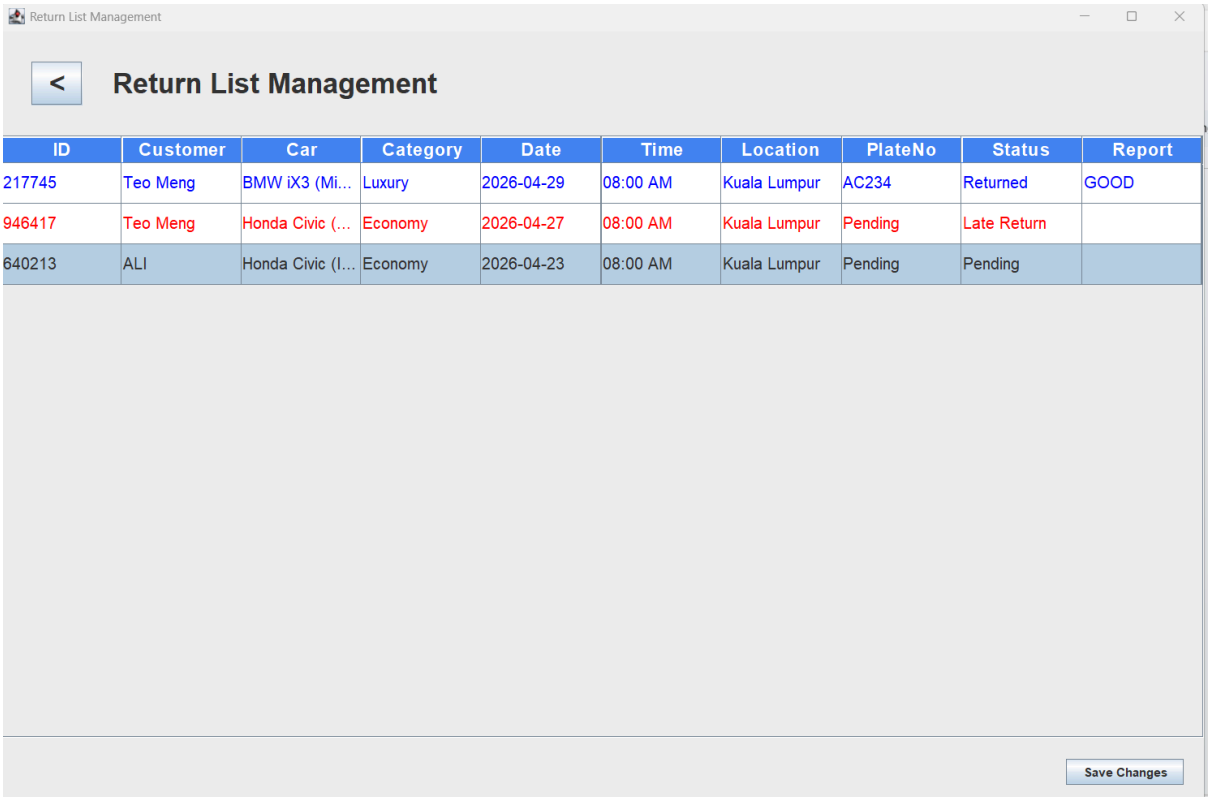


// After staff enter the report



// Staff enter report





// System display message

In the Return List Page, it will display all active bookings that are pending return. Staff can process vehicle returns and calculate the final payment based on rental duration, fuel

condition, and any damages. The system will automatically update the booking status to "Returned" or "Late Returned" based on the return time.

Steps:

1. Staff selects a row.
2. Staff checks vehicle condition, including fuel level and any damages.
3. Staff click the black space to edit it.
4. Staff enter the car report.
5. After entering, staff press the **Save Changes** button to save the report.
6. The system will display “Data Saved Successfully!”

Staff can also click the **Back** button in the upper left corner to return to the **Staff Main Page**.

5.0 Conclusion

In conclusion, the Car Rental Management System provides an effective solution for managing vehicle rental operations. The system helps to organize vehicle information, customer data and booking processes in a structured way. By applying object oriented programming concepts such as encapsulation, inheritance and polymorphism, the system improves code organization, security, and flexibility. In addition, the system reduces manual errors in calculations, improves efficiency in daily operations and provides a better user experience for both customers and staff. Overall, the system is reliable, easy to use and suitable for managing car rental services.

6.0 References

1. SOCAR Malaysia. (n.d.). Car rental service. Retrieved from https://socar.my/consumer

2. Hertz. (n.d.). Car rental services in Kuala Lumpur, Malaysia. Retrieved from https://www.hertz.com/us/en/location/malaysia/kualalumpur/kulc60